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Measurement of corrections to the electron charge misidentification rate in Monte Carlo simulation using the full Run 2 dataset collected with the ATLAS detector

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Abstract

This thesis presents an analysis of the electron charge misidentification (denoted QmisID or charge-flip) rate in exactly two lepton same sign (2LSS) electron channel using Run 2 collision data from the ATLAS experiment at the Large Hadron Collider (LHC). Charge misidentification is an important background contribution in searches for rare Standard Model processes and physics beyond the Standard Model such as in analyses involving same sign leptons. This study focuses on electrons reconstructed with incorrect charge in the 2LSS region, originating primarily from the Z +jets process. We employ the tag-and-probe method to identify and characterize charge-flip events. The tag is chosen such that it targets the electrons with the correctly reconstructed charge while data-driven measurements were performed on the probe electrons. Scale factors were derived from the data and applied to charge-flip MC to improve the modeling of simulated events. The method was validated using dedicated validation regions enriched in charge-flip contribution from other background processes such as $t\bar{t}$ and other top-quark-related background sources.

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Chapter 1

Introduction

The Standard Model (SM) of particle physics provides an exceptionally successful description of fundamental particles and their interactions. However, experimental observations show that it is an incomplete theory of the nature. Therefore, it is desired to search for physics beyond the SM at the highest accessible energies. Yet, to perform searches for new phenomena, it is a necessity that the processes described by the SM (hereafter referred to as background) are understood very well. With this goal in mind, we present an analysis of the QmisID background in the final states with two electrons with the same charge (hereafter referred to as 2lSS region).

The study analyzes the full (2015-2018) Run-2 proton-proton collision data at center of mass energy $\sqrt{s} = 13$ TeV and luminosity $L = 140 \text{ fb}^{-1}$, collected by the ATLAS detector.

The charge misidentification is only relevant for electrons. It is caused by instrumental effects and is difficult to model accurately in simulations. As a result, most physics analyses estimate the QmisID contribution purely using data-driven techniques.

A common approach is to select a sample of two lepton opposite sign (2lOS) events in data and measure per-electron QmisID rates directly in data. Although this approach provides a robust estimate of the charge-flip background, it has limitations such as the statistical precision of the 2lOS sample. Furthermore, it can not be applied to events with an odd number of leptons. Therefore, being able to use a corrected simulation would address these limitations, and this is what this work aims to achieve.

We begin with the theoretical background in chapter 2, introducing the SM and the production of electrons at the LHC. The principles of the charge reconstruction, the origin of same charge di-electron events and classification of relevant background are also discussed.

In chapter 3, a general description of the LHC and the ATLAS detector is provided. We also discuss the tag and probe method and the usage of the boosted decision tree (BDT), used in this study.

Chapter 4 contains the discussion on how various particles are reconstructed in the ATLAS detector. The reconstruction of the electrons and their charge determination is emphasized. Chapter 5, outlines a brief discussion on the generation of the MC simulated samples with various event generators together with a detailed discussion on the various, relevant sources background.

In chapter 6, we present the analysis strategy used to derive the charge-flip corrections in the 2lSS electron region. A charge-flip enriched control region is defined. The prompt and charge-flip electrons are categorized into tag and probe candidates using the output of a BDT discriminant. Finally, we derive the scale factors as a function of the probe electron kinematic properties. Suitable validation regions are also defined involving b -jets and an electron-muon region.

In chapter 7 the results of the charge-flip corrections are presented. Firstly, the scale factors derived in the charge-flip control region are discussed together with their dependence on the charge-flip electron kinematics. The impact of the corrections on the modeling of charge-flip events is evalu-

ated by comparing data to MC agreement before and after applying the corrections. To assess the applicability of the method, the results obtained in validation regions are also presented.

Chapter 2

Theoretical background

2.1 The Standard Model

The Standard Model (SM) of particle physics is the theoretical framework that describes the fundamental constituents of matter and their interactions through the electromagnetic, weak and strong interactions. The particles are classified into fermions and bosons. Fermions include six quarks and six leptons (e , μ , τ) arranged in three generations as shown in figure 2.1. The interactions between the fermions are mediated by the gauge bosons (γ , Z , $W^\pm$ and gluons). The discovery of the Higgs boson in 2012 completed the particle content of the SM which confirmed the mechanism responsible for generation of masses of the Z and $W^\pm$ gauge bosons.

Although the SM is successful, there are reasons to expect new phenomena at the LHC and searching for these requires a detailed understanding of the SM processes. We are interested in the analysis of final states with two electrons with the same charge (2ISS). One such background that can enter this region is via charge-misidentification, the focus of the thesis.

Standard Model of Elementary Particles									
THREE GENERATIONS OF MATTER (FERMIONS)						INTERACTIONS / FORCE CARRIERS (BOSONS)			
QUARKS	I		II		III		GAUGE BOSONS (vector bosons)		SCALAR BOSON (scalar particle)
	mass $\approx 2.16 \text{ MeV}/c^2$	u up	$\approx 1.273 \text{ GeV}/c^2$	c charm	$\approx 172.57 \text{ GeV}/c^2$	t top	0 0 1	g gluon	$\approx 125.2 \text{ GeV}/c^2$ 0 0 H higgs
	charge $+2/3$		$+2/3$		$+2/3$		0 0 1	γ photon	
	spin $1/2$		$1/2$		$1/2$		$\approx 91.188 \text{ GeV}/c^2$ 0 1	Z Z boson	
		$\approx 4.7 \text{ MeV}/c^2$	d down	$\approx 93.5 \text{ MeV}/c^2$	s strange	$\approx 4.183 \text{ GeV}/c^2$	b bottom	$\approx 80.3692 \text{ GeV}/c^2$ +1 1	W^+ W ⁺ boson
		$-1/3$		$-1/3$		$-1/3$		$\approx 80.3692 \text{ GeV}/c^2$ -1 1	W^- W ⁻ boson
LEPTONS		$\approx 0.511 \text{ MeV}/c^2$	e electron	$\approx 105.66 \text{ MeV}/c^2$	μ muon	$\approx 1.77693 \text{ GeV}/c^2$	τ tau		
		-1		-1		-1			
		$1/2$		$1/2$		$1/2$			
		$< 0.8 \text{ eV}/c^2$	ν_e electron neutrino	$< 0.17 \text{ MeV}/c^2$	ν_μ muon neutrino	$< 18.2 \text{ MeV}/c^2$	ν_τ tau neutrino		
		0		0		0			
		$1/2$		$1/2$		$1/2$			
- Masses are given in energy units (eV/c^2). - For neutrinos, the listed mass is an upper limit. - Spin is given in units of $\hbar$.						- Quarks (6) - Leptons (6) - Gauge bosons (5) - Scalar boson (1)			
						- mass: Rest mass of the particle (energy) - charge: Electric charge in units of the elementary charge (e) - spin: Intrinsic angular momentum (in units of $\hbar$)			

Figure 2.1: Elementary particle content of the SM. The fermions (six quarks and six leptons) are arranged into three generations. The interactions between these particles are mediated by the gauge bosons associated with the strong, electromagnetic, and weak forces, while the Higgs boson is responsible for electroweak symmetry breaking and the generation of particle masses.

2.2 Electrons production

In proton-proton (pp) collisions, electrons are produced through several SM processes. The dominant sources include the decays of electroweak gauge bosons, such as $Z \rightarrow e^+e^-$ and $W^\pm \rightarrow e^\pm\nu_e$

along with top-quark pair and diboson production processes. Among these the decay of the Z boson into electron-positron pair provides a clean experimental signature due to the well known mass of the particle and relatively low background contamination.

In proton-proton collisions, the electron pairs with opposite charge are more likely to be produced than electron pairs with the same charge. Therefore, any mechanism that causes the charge of an electron to be reconstructed incorrectly can transform an opposite sign event into a same sign event. This makes it an important background in our study of the analysis of two electrons with same charge.

Determination of charge

The measurement of the electric charge of an electron is determined by its motion through the strong solenoidal magnetic field of the ATLAS detector. A charged particle traversing the magnetic field experiences the Lorentz force, given as in eq. 2.1:

$$\vec{F} = q(\vec{v} \times \vec{B}), \quad (2.1)$$

where, q corresponds to the charge of an electron, $\vec{v}$ is the velocity vector of an electron and $\vec{B}$ is the magnetic field intensity. The Lorentz force causes the trajectory of electron to bend. The direction of the curvature determines the sign of the electric charge. Additionally, the magnitude of this curvature also determines the 4-momentum of the particle.

In the ATLAS experiment, the trajectories of electrons are reconstructed using the measurement from the Inner Detector, discussed in detail in section 4. However, bremsstrahlung paired with photon conversion can result in a mis-reconstructed track curvature, which in turn may cause the charge of the electron to be reconstructed incorrectly.

Di-electron events with same charge

Final states of two leptons with the same charge are rare within the SM. Background processes contributing to the same sign dilepton events are divided into two categories:

- **Irreducible backgrounds:** The SM processes that genuinely produce final states with two leptons with same electric charge are called irreducible backgrounds. These backgrounds arise from the processes (prompt sources), such as $t\bar{t}W$ and $W^\pm W^\pm$. They can be estimated reliably with MC simulations.
- **Reducible backgrounds:** These backgrounds arise from sources different from prompt processes. They mainly occur due to instrumental effects, particle mis-reconstruction and non-prompt sources such as jets misidentified as electrons or, non-isolated electrons from b/c -hadron decays. QmisID events are one of such reducible backgrounds.

2.3 Tag-and-probe method

The tag-and-probe method [1] is a data-driven technique widely used in collider experiments such as the ATLAS experiment to measure the reconstruction, identification, isolation, trigger and charge identification efficiencies directly from the collision data. The method exploits the decay of well understood resonances, e.g. $Z \rightarrow e^-e^+$ or $Z \rightarrow \mu^-\mu^+$ events. This provides a clean and unbiased sample of leptons. In this approach, one lepton candidate selected as *tag*, is required to satisfy e.g. strict identification and isolation requirements. This ensures a pure sample for the tag selection. The second electron candidate, called *probe*, is then used to evaluate the performance of the detector or reconstruction algorithm under study. The tag-and-probe method allows scale factor measurements directly from data and MC simulations.

In the context of this thesis, the method is applied using $Z \rightarrow e^-e^+$ events with the di-electron invariant mass required to be consistent with Z boson mass.

Chapter 3

Experimental setup

3.1 LHC

The Large Hadron Collider (LHC) is the world's largest and most powerful particle accelerator, located at European Organisation for Nuclear Research (CERN) near Geneva, Switzerland. It is installed in a circular underground tunnel of approximately 27 km in circumference at a depth of about 100 m below the surface. The LHC is designed to accelerate and collide proton beams at very high center of mass energies to understand the fundamental constituents of matter and their interactions.

Inside the accelerator, two counter rotating proton beams are accelerated using superconducting radio-frequency (RF) cavities and guided by strong superconducting dipole magnets. The beams collide at four interaction points where major experiments such as ATLAS (A Toroidal LHC ApparatuS), CMS (Compact Muon Solenoid), ALICE (A Large Ion Collider Experiment) and LHCb (Large Hadron Collider beauty) are located. The high collision energies achieved at LHC allow physicists to investigate phenomena such as Higgs boson, top physics, electroweak interactions and possible physics beyond the Standard Model.

The LHC began its operations in 2008 and achieved its most important milestone with the discovery of Higgs boson. In June 2026, after Run-3, it will be further upgraded for the new for High-Luminosity LHC (HL-LHC) phase.

3.2 ATLAS detector

The ATLAS experiment [2] [3], (figure 3.1) is one of the two general purpose particle detectors at the LHC with a forward-backward symmetric, cylindrical geometry and an almost 4π coverage in solid angle. It consists of an inner detector (ID) surrounded by a superconducting solenoid, electromagnetic and hadronic calorimeters (ECAL and HCAL), and a muon spectrometer incorporating large superconducting air-core toroidal magnets. It is designed to identify a wide variety of particles and measure their energies and momenta.

Inner Detector (ID)

The inner tracking detector is surrounded by a 2T solenoidal magnetic field and helps identify tracks of charged particles in the $|\eta| < 2.5$ pseudorapidity range. It has three main components, namely: the silicon pixel detector, semiconductor tracker (SCT) and transition radiation tracker (TRT).

The pixel detector has 92M readout channels to provide high granular spatial tracking of charged particles very close to the interaction point. It enables a precise vertex reconstruction and efficient identification of short lived particles such as b-hadrons and tau leptons. The SCT surrounds the pixel detector and has more than 6M readout channels, allowing the precise momentum reconstruction.

It is followed by the TRT, which is the outermost part of the ATLAS Inner Detector and is designed for both charged-particle tracking and electron identification. The TRT provides 30 precision points per track with spatial resolution of $130\mu\text{m}$, significantly improving the momentum

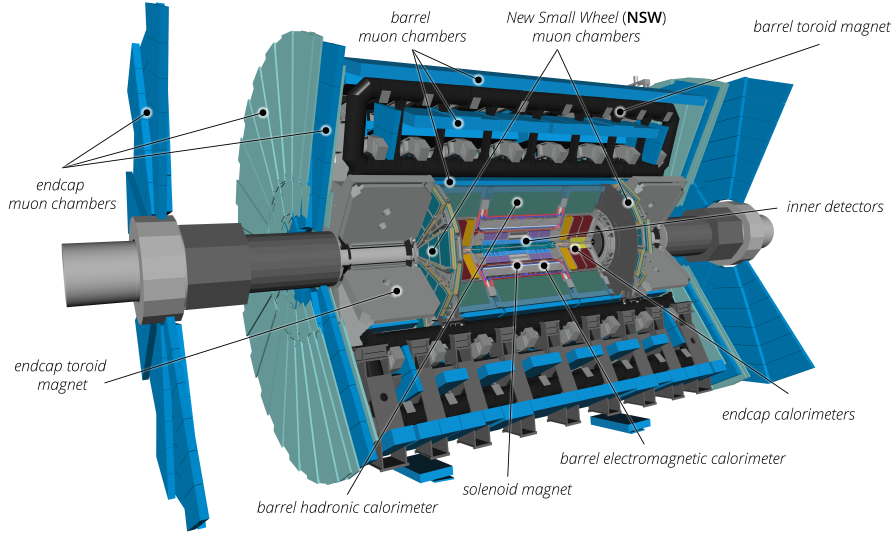


Figure 3.1: A schematic diagram of the ATLAS detector (Image: ATLAS experiment/CERN)

measurement of the particles. It operates in the pseudorapidity range $|\eta| < 2.0$.

Calorimeters

The calorimeter system at ATLAS is designed to measure the energy of the particles produced in the proton-proton collisions. It surrounds the ID, absorbs particles and then their energy is reconstructed from the resulting particle showers. The calorimeter system is divided into two parts: electromagnetic and hadronic calorimeters. The electromagnetic calorimeter (ECAL) is optimized to detect and measure electrons and photons with very high precision. The ECAL is a liquid-argon (LAr) sampling calorimeter with lead absorbers. It is divided into barrel and endcap regions covering central and forward regions respectively. When an energetic electron or photon enters the detector, it produces an electromagnetic shower through bremsstrahlung and pair production processes. The lead absorber initiates the shower, while the liquid argon acts as the active medium where ionization signals are collected. The ECAL geometry is optimized such that it provides full azimuthal coverage and fast signal collection without gaps in the detector acceptance.

Unlike the ECAL, the hadronic calorimeter measures the energy deposited by strongly interacting particles such as hadrons. It is composed of: tile calorimeter, hadronic endcap calorimeter (HEC) and forward calorimeter (FCal). The Tile Calorimeter covers the central region of the detector and uses steel absorber plates interleaved with scintillating tiles as the active medium. When hadrons interact with the absorber material, they produce hadronic showers and the resulting scintillation light is collected and converted into electrical signals for their energy reconstruction. The HEC and FCal cover the forward regions where the particle flux is high. The HEC employs liquid argon (LAr) as the active medium with copper absorbers, while the FCal uses dense absorber material to withstand the radiation hard environment closer to the beam-pipe.

Muon spectrometer

The muon spectrometer [4] is the outermost component of the ATLAS detector and is specially designed to detect and measure muons. Since muons interact only weakly with matter, they can penetrate the inner detector and calorimeter with relatively small energy loss.

It operates inside a large air-core toroidal magnetic field produced by superconducting barrel and endcap toroid magnets. This provides precise transverse momentum measurement of the muon over the pseudorapidity range in the barrel of $|\eta| < 1.05$ and in the two endcaps of $1.05 < |\eta| < 2.7$.

Chapter 4

Object reconstruction

4.1 Particles reconstruction and identification

Particles produced in proton–proton collisions are reconstructed in the ATLAS detector using information from its different subdetectors. As particles traverse the detector, they interact with the detector material in different ways depending on their properties such as charge, mass, and interaction type. Charged particles leave tracks in the Inner Detector, allowing the measurement of their momentum and charge through their curvature in the magnetic field. Electrons and photons deposit their energy primarily in the electromagnetic calorimeter, while hadrons produce showers mainly in the hadronic calorimeter. Muons penetrate through the calorimeters and are reconstructed by the information in the Muon Spectrometer and the ID. By combining information from these detector systems, ATLAS reconstructs and identifies the particles produced in collision events. This chapter focuses on the reconstruction, identification, and calibration processes of electrons.

4.2 Electron reconstruction and identification

Electron reconstruction [5] in the ATLAS detector is based on combining information from the Inner Detector and the electromagnetic calorimeter. Electrons produced in proton–proton collisions first traverse the tracking detectors, where they leave hits, and then deposit all of their energy in the electromagnetic calorimeter through the development of an electromagnetic shower. The reconstruction procedure therefore relies on identifying electromagnetic energy clusters in the calorimeter and matching them to charged-particle tracks reconstructed in the ID.

Using the 3×5 towers, localized energy deposits are searched in the ECAL via sliding-window algorithm, cluster candidates with energy $E_T > 2.5$ GeV are retained. It is seen that the reconstruction efficiency increases with increasing energy deposits in the electromagnetic calorimeter i.e. 65% at 4.5 GeV to more than 99% above 15 GeV.

After cluster reconstruction, charged particle tracks are reconstructed in the Inner Detector using hits from the Silicon detectors. In the pixel detector, each cluster equates to one 3-D measurement, while in the SCT, clusters from both stereo views of a strip layer must be combined to obtain a 3-D measurement. Track seeds are formed via three 3-D points in Silicon detector layers and then, track reconstruction proceeds in three steps which are pattern recognition, ambiguity resolution and TRT extension[6].

After reconstruction, electron candidates are subjected to identification and isolation requirements. They are identified using the information from ID, ECAL such as shower-shape variables, and track quality. Isolation requirements ensure that electrons are well separated from nearby tracks and energy deposits to reduce contamination from jets and non-prompt electrons. Working points are defined based on the desired identification efficiency and isolation.

A major difficulty in the electron reconstruction is the presence of bremsstrahlung in which an electron radiates a photon that can interact with the detector material triggering pair-production. As the electron travels the detector material it loses energy and due to this the track momentum

measurement of the electron is affected. This issue is addressed using an optimized Gaussian Sum filter (GSF) [7], to account for the energy loss of charged particles in the material.

Electrons are matched to the primary vertex (PV) by requiring d_0 is satisfying $|d_0|/\sigma_{d_0} < 5$ (where, d_0 is transverse impact parameter and σ_{d_0} is its measured uncertainty), and $|z_0 \sin\theta| < 0.5$ mm (where, z_0 is the measured distance from the PV to track along the beam-pipe).

In this study, electrons are required to satisfy $p_T > 20$ GeV and $|\eta| < 2.47$ and the transition region ($1.37 < \eta < 1.52$) between barrel and endcaps of the ECAL is excluded. Electron candidates can be reconstructed with different identification and isolation working points, categorized as Loose, Medium and Tight. For this analysis, electrons are required to satisfy the Tight identification and Tight isolation criteria.

Electron charge reconstruction and QmisID events

The charge of an electron candidate can be misidentified due to mismeasurement of the curvature of its associated track. At high p_T , tracks become straighter which could lead to an incorrect charge determination of the electron.

However, the dominant source of electron charge misidentification is due to bremsstrahlung [8]. The electron can radiate a photon which subsequently produces an electron-positron pair. This can lead to the association of the wrong track of an electron candidate as shown in fig. 4.2.

The cross section of the bremsstrahlung increases with p_T and η . It increases at large η as the electron traverses more the detector material.

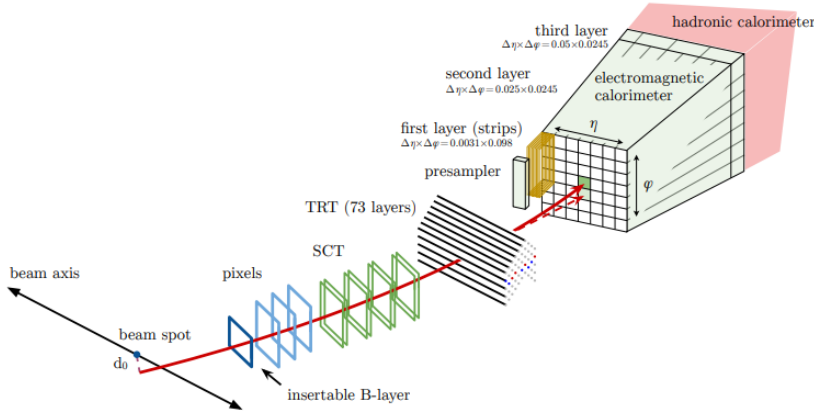


Figure 4.1: An illustration of path of an electron through the ID and calorimeter system of the ATLAS detector [5].

Boosted Decision Tree (BDT)

A Boosted Decision Tree (BDT) [9] is a supervised machine learning algorithm commonly used in high energy physics. A BDT exploits the discriminating power of multiple variables simultaneously to separate different classes of events. A decision tree consists of a sequence of binary selections that recursively partition the feature space into regions enriched with signal or background events. A single tree can provide event classification, but its performance is limited by statistical fluctuations and sensitivity to the training sample. These limitations are overcome by using a boosting method. This technique combines the predictions of many sequentially trained decision trees. The events that are incorrectly classified in one tree are given more weight in the subsequent tree. The final BDT discriminant is obtained from a weighted combination of all trees in the ensemble known as BDT score.

In this study, a BDT called Electron Charge Identifier Selector (ECIDS) is used to distinguish correctly reconstructed electrons from charge-flip electrons using the variables associated to the electron track and the calorimeter cluster. A "Loose" ECIDS working point is defined to reject charge-flip electrons, by requiring their ECIDS score to be ≥ -0.37 .

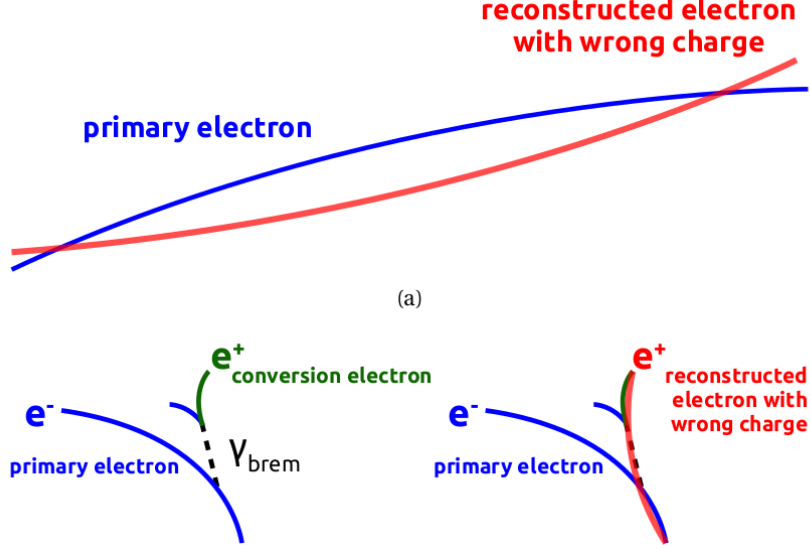


Figure 4.2: The two different sources of electron charge misreconstruction: straight high- p_T track (top) and Bremsstrahlung (bottom).

4.3 Muon

The Muon candidates [10] are reconstructed by combining information from both muon spectrometer and ID. In this work, muons are required to satisfy $p_T > 20$ GeV and $|\eta| < 2.5$. A 'Tight' working point is defined and used for muons as well. Like electrons, muons are also matched to the PV by requiring that d_0 satisfies $|d_0|/\sigma_{d_0} < 3$, and $|z_0 \sin\theta| < 0.5$ mm. Reconstructed muons are calibrated using the data acquired via the processes, $Z \rightarrow \mu^+\mu^-$ and $J/\Psi \rightarrow \mu^+\mu^-$. By combining tracks from both the MS and ID, ATLAS achieves a great performance of muons, contributing to their accurate and precise identification. Muons in study, are only considered in the validation regions.

4.4 Jets

Strongly interacting particles produced after each proton-proton (pp) collision initiate parton showers (radiating gluons and quark-antiquark pairs). These partons can not be free and, via the process of hadronisation, form hadrons. The formed hadrons have a certain measurable lifetime after which they decay to lighter particles and form a collimated spray of particles in the calorimeter, called jet [11]. Jets are reconstructed from tracks and clusters of energy in the hadronic calorimeter using the anti- k_t algorithm with a radius parameter of $R=0.4$. Jets originating from b -quarks (b-jets) are identified using a multivariate b -tagging algorithms. This exploits the lifetime and kinematic properties of b/c -hadron decays.

In this work, only b-tagged jets that pass the threshold $p_T > 20$ GeV and $|\eta| < 2.5$ are selected. For b-jets, several Working Points (WPs) are defined based on the efficiency for b -tagging a jet. In this analysis, jets at 85% efficiency WP are considered.

4.5 Missing transverse momentum

Given the conservation of the total momentum in the transverse plane (perpendicular to the beam direction), the missing transverse momentum is defined as the imbalance of the momentum in such

plane from particles that fail to be detected, e.g. neutrinos (ν).

Thus, the quantity p_T^{miss} is calculated as the negative vector sum of the p_T of all the other selected and calibrated objects in the event, including also a term to account for the momentum from soft tracks in the event that are not associated with any of the other reconstructed objects. This soft term is measured from ID tracks matched to the selected PV [\[12\]](#).

Chapter 5

Data and Monte Carlo simulations

Given the impossibility to distinguish which processes contribute to data collected from pp collisions, Monte Carlo (MC) simulations are used to model both physics processes and the response of the detectors.

The MC simulation of collision events is performed via dedicated event generators. They model the collision and different stages of particle interaction. Sherpa and Powheg are some of the Matrix element generators for the hard-scattering process. The processes such as Parton Shower (PS) and hadronisation are often simulated with Pythia or HERWIG. Additional pp collisions in the same or nearby bunch crossings (pile-up), are also simulated using e.g. Pythia [13].

The particles generated are subjected then to the full ATLAS detector simulation [14] using Geant4 [15]. Some of the background processes use a fast-simulation [16] that replaces the full Geant4 treatment of the calorimeter response with a parameterization of the shower characteristics, reducing the computational requirements while maintaining an acceptable performance.

The data sample collected by the ATLAS Collaboration at $\sqrt{s} = 13$ TeV for full Run-2 (2015-2018) is used in the current analysis. For this dataset the integrated luminosity after beam and data quality requirements is 140 fb^{-1} . Moreover, MC samples are normalized based on their respective cross-sections and are generated to the highest available order.

5.1 Background description

The term "Background" refers to the processes other than the targeted one which also contribute to the region of interest. It can be classified into two categories *reducible* and *irreducible* backgrounds as outlined in section 2.2

Background events can be categorized using their truth information into those containing only prompt leptons, and those featuring non-prompt leptons from different sources. Prompt events are those that originate directly from the primary interaction point after the immediate decay of particles produced in pp collisions. Whereas the non prompt events arise from sources that do not involve primary particle interactions. However, these events can also result from instrumental effects such as detector noise or interactions with detector material which may not be well modeled in MC simulations. Particles that originate slightly away from the primary vertex (PV), known as secondary vertex, also constitute non prompt events. E.g. Particles arising from the decay of b -hadrons originate from secondary vertex. In this work, a number of MC samples contributing only prompt leptons to the 2lSS final state such as diboson (VV) are used. Others are grouped in the "Top" and "Other" categories. The 'Top' category involves the processes that have at least one top-quark such as $t\bar{t}W$ except $t\bar{t}$ which is considered separately. The 'Other' category contains the remaining processes that do not have any top-quarks such as $V\gamma$. In either cases, they hold minimal relevance in the analysis.

There are several other reducible backgrounds that contribute to 2lSS events. These are heavy-flavor (HFe) and light-flavor (LFe) electrons originating from b/c and $u/d/s$ hadronic decays respectively. Similarly, HFmu and LFmu correspond to muons from heavy-flavor and light-flavor

decays, respectively. Photon conversions (Conv) are e.g. promptly-produced photons which subsequently are converted to electrons in the detector material. "Unknown" events are those which can not be attributed to any of the truth type due to the lack or corruption of truth information in MC simulations. The region selection is optimized to enhance the charge-flip purity while reducing the impact of these additional reducible backgrounds.

Chapter 6

Analysis strategy

6.1 Event pre-selection

A set of pre-selection criteria is required to ensure a well reconstructed event sample of events with exactly two electrons with the same electric charge, corresponding to a total charge of $Q = \pm 2$. Reconstructed muons or tau leptons are vetoed. The electrons are required to satisfy $p_T^0 > 27$ GeV and $p_T^1 > 20$ GeV respectively (where the superscripts 0 and 1 refer to their p_T ordering). They are also required to pass Tight likelihood identification and Tight isolation criteria. Events containing b -tagged jets are vetoed to avoid contamination from heavy flavor electrons. The resulting event sample serves as the starting point for the definition of the charge-flip control region. For the pre-selected event sample, the di-electron mass distribution is shown in fig. 6.1.

6.2 Charge-flip control region

Having defined the pre-selection criteria, the charge-flip control region is then defined to enhance the contribution of events with one of the electrons having a misreconstructed charge. The idea is to select a region where charge-flip events dominate and then to measure the corrections to charge-flip MC events by comparing the total MC background to data.

The major contribution of charge-flip comes from the on-shell Z +jets process as seen in fig. 6.1. Therefore, to acquire the purest possible charge-flip control region, the di-electron invariant mass around the Z mass peak (i.e. $76.6 < m_{ll} < 101.2$ GeV) is selected as shown in fig. 6.2. Fig. 6.2(b) shows the contribution of various processes in this region grouped based on their truth lepton origin. From this it can be seen that the purity of charge-flip events is $\sim 80\%$, which justifies the chosen event selection. The event selection cuts are summarized in table 6.1.

Simulated electrons are characterized based on their truth types. The truth-type categories for electrons are described in table 6.2. Electrons with truth-type 1 are classified as prompt and those with truth-type 2 are classified as charge-flip electrons.

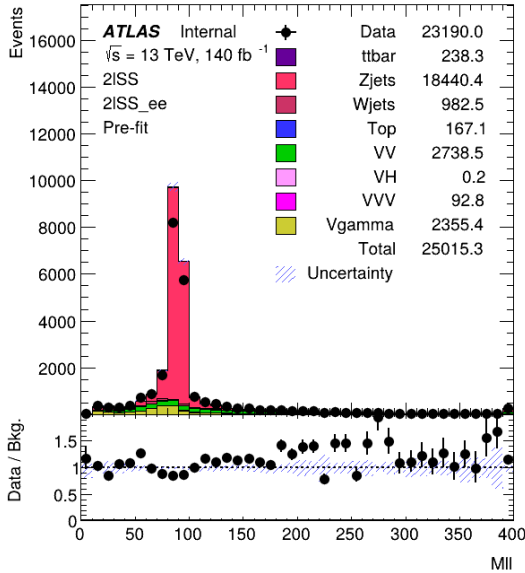
Following the truth-based classification, it is required to target prompt electrons as *tag* and charge-flip electrons as *probe*. Such distinction has to be achieved only through reconstructed quantities, since truth information is not available in data. For this purpose several variables can be considered. Among them, the ECIDS score provides the best discrimination for the tag-and-probe characterization of the prompt and charge-flip electrons. This is illustrated in fig. 6.3. A clear separation is seen between prompt and charge-flip electrons at higher ECIDS score. Figure 6.4 shows the next-best discriminator, the pseudorapidity η , for comparison.

Given the stronger separation achieved by the ECIDS score, the tag electron is defined as the electron with the highest ECIDS score in the event. This choice maximizes the probability that the tag corresponds to prompt electrons.

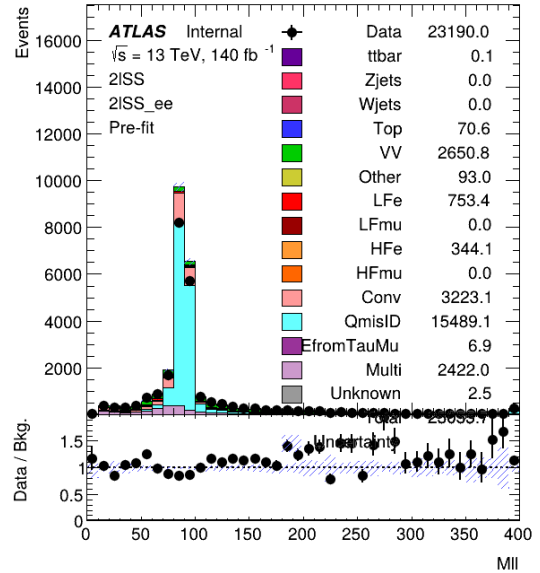
The validity of this tag definition can be assessed using the truth-type information of the selected tag-and-probe electrons. Figure 6.5 shows the truth-type distributions of the selected electrons. The chosen tag electron is dominated by prompt electrons with a small contribution from photon conversions (fig. 6.5(a)), while the probe sample is predominantly composed of charge-flip electron

Table 6.1: The selection criteria for the control and validation region.

Selection criteria	Control Region	Validation Regions
ee channel		
N_e	2	2
$ \sum Q_{ee} $	2	2
Electron charge identification (ECIDS) WP	<i>Loose</i>	<i>Loose</i>
Electron p_T threshold	$p_T^{e_0, e_1} > 27, 20 \text{ GeV}$	$p_T^{e_0, e_1} > 27, 20 \text{ GeV}$
<i>ID</i> and Isolation WP	TightLH, Tight	TightLH, Tight
Di-lepton invariant mass ($m_{\ell\ell}$)	$76.6 < m_{ll} < 101.2 \text{ GeV}$	–
$N_{b\text{-jets}}$ (@85% WP)	0	$= 1 \text{ or } \geq 2$
Missing energy (E_T^{miss})	–	$< 50 \text{ GeV}$
AmbiguityType and DFCAA	–	$= 0 \text{ and } \leq 0$
$e\mu$ channel		
N_e, N_μ	–	1,1
$ \sum Q_{e\mu} $	–	2
Electron charge identification (ECIDS) WP	–	<i>Loose</i>
Electron/muon p_T threshold	–	$p_T^{e_0, \mu_1} > 27, 20 \text{ GeV}$
<i>ID</i> and Isolation	–	TightLH, Tight
$N_{b\text{-jets}}$ (@85% WP)	–	≥ 2



(a)



(b)

 Figure 6.1: Di-electron invariant mass ($m_{\ell\ell}$) distributions for electrons in 2lSS channel. Plots (a) and (b) show the full 2lSS event sample categorized by different processes and electron truth origin, respectively.

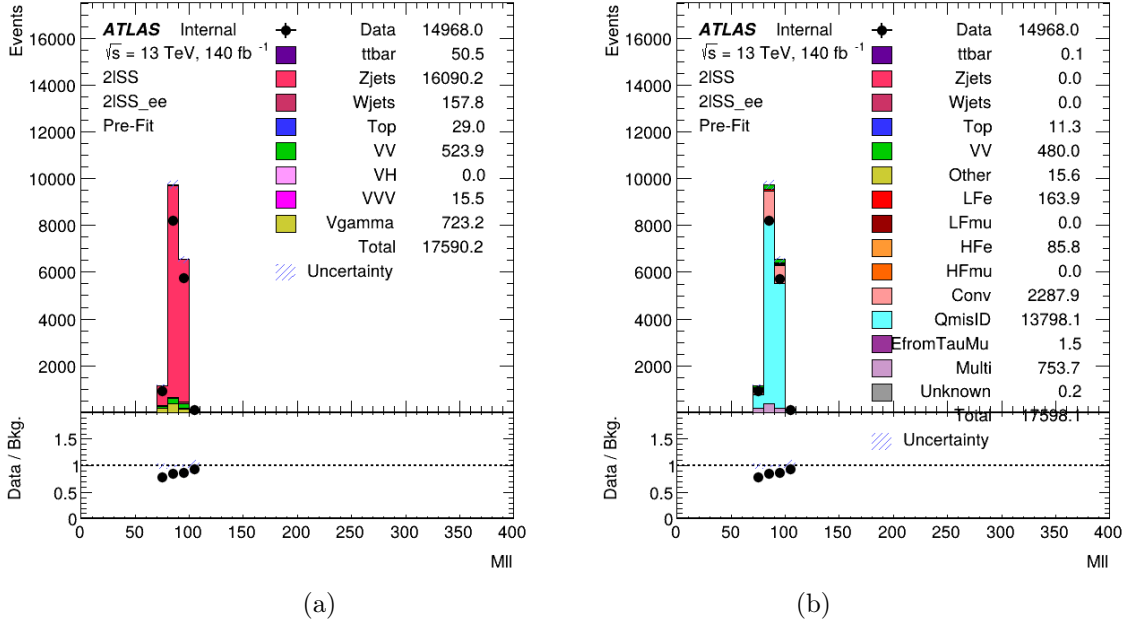


Figure 6.2: Di-electron invariant mass ($m_{\ell\ell}$) distributions for electrons in 2lSS channel. Plots (a) and (b) show the distributions after selecting the Z-boson mass peak region.

(fig. 6.5(b)). The efficiency of selecting the prompt electron as tag is $\sim 90\%$. The remaining $\sim 10\%$ is taken as a source of systematic uncertainty for the measured corrections

After the tag selection, the p_T and η distributions of the probe are examined in fig. 6.6. MC provides already a good description for both distributions, reproducing overall the shape and normalization observed in data.

Measurement of Scale Factors

The ratio between data and total MC background of probe electrons shown in figures 6.6(a) and 6.6(b), represent the level of mismodeling of charge-flip MC simulation and, thus, the size of the corrections that should be considered. The fact that there is a different dependency for the data/MC ratio versus p_T and η implies the need to evaluate the corrections versus the probe electron p_T and η simultaneously. For this purpose it is necessary to arrange data and MC events in a 2D histogram as shown in fig. 6.7 for data. The p_T axis is binned so that the number of events in each bin is such that the relative statistical uncertainty in data remains under 30%. The $|\eta|$ axis is binned in segments of 0.5 (referred to as $|\eta|$ slices), keeping the detector geometry in mind. Following this, the p_T distributions for each $|\eta|$ slice are analyzed separately. Charge-flip electron corrections are then derived using data and MC predictions in every bin. Scale factors are defined by the ratio of charge-flip rate in data to the charge-flip rate in MC. Charge-flip rate in data is estimated by subtracting from data the MC yield of non charge-flip events.

The expression to measure scale factors is given below:

$$SF(i, j) = \frac{N_{\text{Data}}(i, j) - N_{\text{other MC}}(i, j)}{N_{\text{QmisID}}(i, j)}, \quad (6.1)$$

where, $N_{\text{Data}}(i, j)$ corresponds to the data values for the probe electron in the i^{th} p_T bin and j^{th} $|\eta|$ bin, while $N_{\text{other MC}}(i, j)$ and $N_{\text{QmisID}}(i, j)$ are the number of other/non charge-flip and charge-flip MC events in the same bin, respectively. For different values of scale factors, the MC yields either underestimate or overestimate or accurately predict the data as illustrated in equation 6.2:

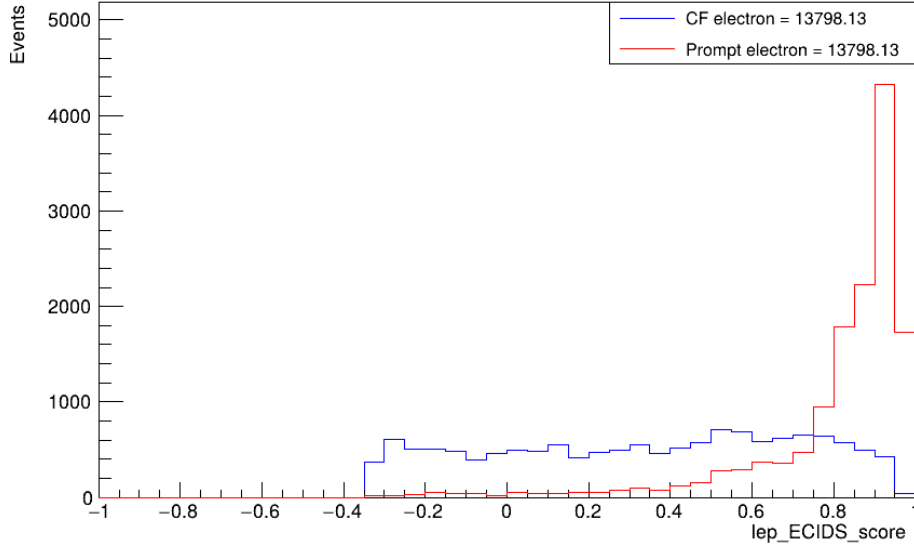


Figure 6.3: Distribution of the ECIDS score for prompt and charge-flip electrons in Z +jets MC. Prompt electrons (red) are concentrated at large ECIDS values, while charge-flip electrons (blue) populate lower ECIDS regions, demonstrating the discriminating power of the ECIDS score for the tag-and-probe selection.

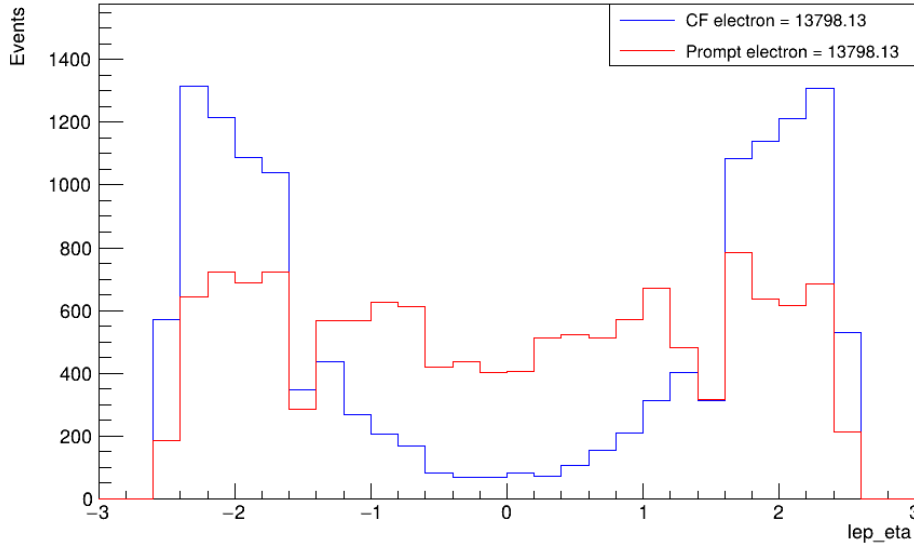
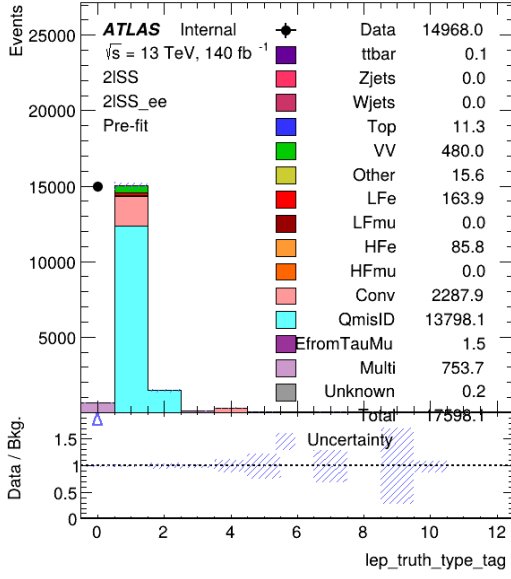
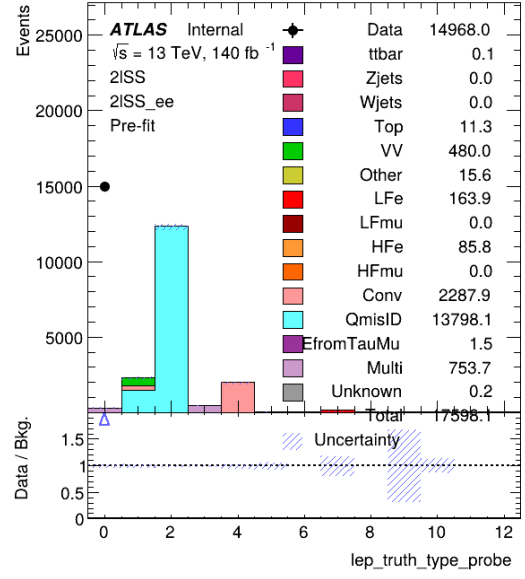


Figure 6.4: Distribution of pseudorapidity (η) for prompt and charge-flip electrons in Z +jets MC. Prompt electrons (red) are mostly uniformly distributed, while charge-flip electrons (blue) are more abundant at larger values of ($|\eta|$), reflecting the increased probability of charge misidentification in the forward detector regions.

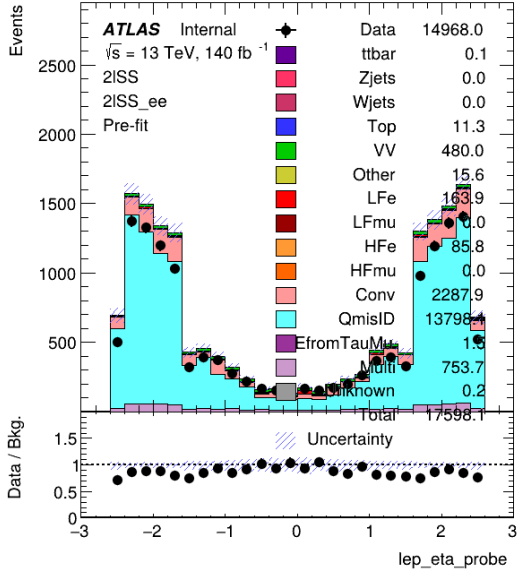


(a)

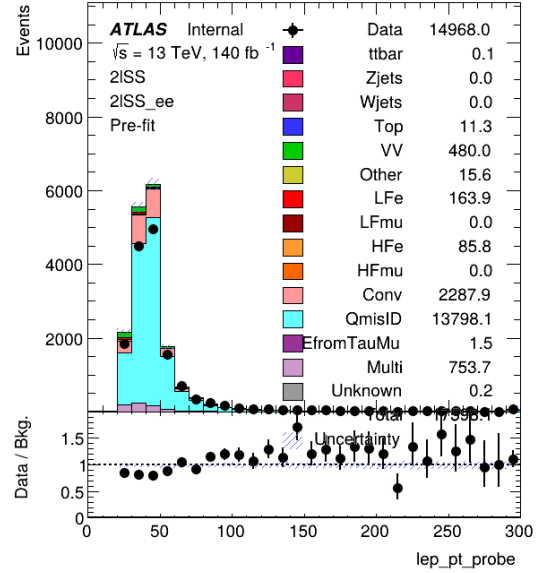


(b)

Figure 6.5: Truth-type distributions of tag (a) and probe (b) electron candidates in charge-flip control region. Data in both is concentrated at zero as it does not have truth information.



(a)



(b)

Figure 6.6: Pseudorapidity (η) (a) and transverse momentum (p_T) (b) distributions of selected probe electrons in the charge-flip control region.

$$SF \rightarrow \begin{cases} > 1, & \text{MC underestimates the charge-flip yield,} \\ = 1, & \text{MC accurately describes the charge-flip yield,} \\ < 1, & \text{MC overestimates the charge-flip yield.} \end{cases} \quad (6.2)$$

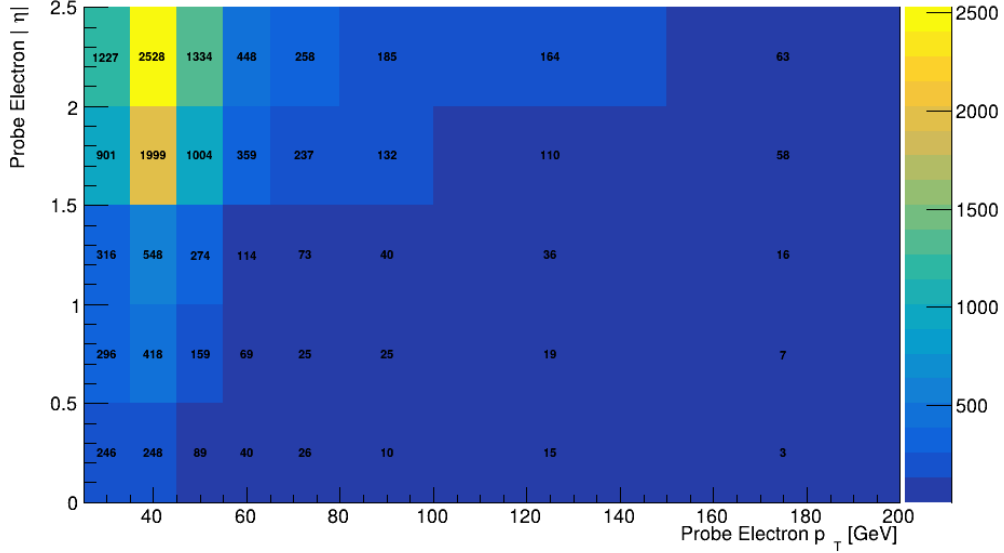


Figure 6.7: Two-dimensional histogram of probe-electron transverse momentum (p_T) and pseudo-rapidity ($|\eta|$) in data. The binning is chosen to maintain statistical uncertainties below 30% and to accommodate the crack-veto region ($1.37 < |\eta| < 1.52$). The numbers shown correspond to the data yields in each $(|\eta|, p_T)$ bin.

The uncertainties for the scale factors are measured based on the standard error propagation formula [17]. The expression for measuring the uncertainties is expressed as:

$$\sigma_{SF} = \frac{1}{N_{QmisID}} \sqrt{\sigma_{Data}^2 + \sigma_{Total MC}^2 + \left(\frac{N_{Data} - N_{Total MC}}{N_{QmisID}} \right)^2 \sigma_{QmisID}^2}, \quad (6.3)$$

It should be mentioned that since the charge-flip yield is roughly equal to the total background yield, the uncertainties for the charge-flip yield are approximated by that of the total yield. Finally, since the data follows the Poisson distribution, the uncertainty is given by the square root of the corresponding data yield.

6.3 Validation regions

Validation regions (VR) defined to be orthogonal to the control region and are used to further verify the charge-flip background correction procedure. Several VRs involving b -jets are considered in the current analysis. Note that allowing b -jets increases the contributions of processes related to top-quark processes. In order to suppress these backgrounds an additional selection cut based on low missing transverse momentum (E_T^{miss}) is applied. It should be mentioned that the di-electron invariant mass requirement is dropped in the $e\mu$ region as the contribution of Z +jets events is already suppressed compared to the corresponding ee region. The VRs are thus defined as follows.

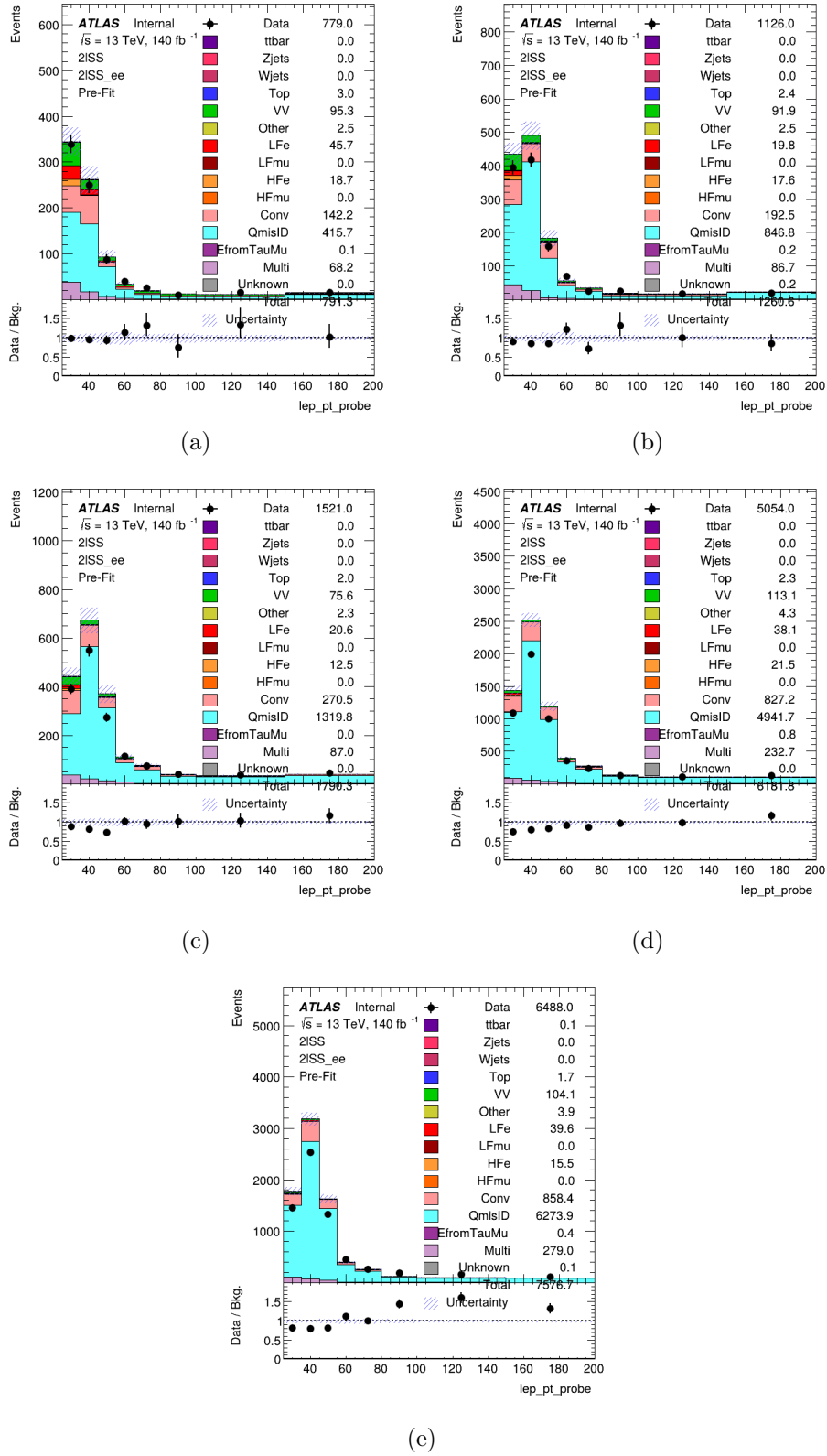


Figure 6.8: Transverse momentum (p_T) distribution of selected probe electrons for different $|\eta|$ slices. Panel (a)-(e) correspond to the ranges $(0 < |\eta| < 0.5)$, $(0.5 < |\eta| < 1.0)$, $(1.0 < |\eta| < 1.5)$, $(1.5 < |\eta| < 2.0)$, and $(2.0 < |\eta| < 2.5)$, respectively.

Table 6.2: Truth-type categories.

Background source	Acronym	Truth-type
Unknown origin	Unknown	0
Prompt electron	Prompt	1
Charge-misidentified electron	ChargeFlipEl	2
Photon conversion	PhotonConv	3
Final-state-radiation photon conversion	FSRphConv	4
Heavy-flavour electron	HeavyFlavourEl	5
Heavy-flavour muon	HeavyFlavourMu	6
Light-flavour electron	LightFlavourEl	7
Light-flavour muon	LightFlavourMu	8
Electron from muon decay	EleFromMu	9
Lepton from τ decay	TauDecay	10
Charge-misidentified muon	ChargeFlipMu	12

ee region:

- (a) Control-region selection with the b -jet veto replaced by a requirement of exactly one b -tagged jet at the 85% efficiency WP + $E_T^{miss} < 50$ GeV.
- (b) Control-region selection with the b -jet veto replaced by a requirement of at least two b -tagged jet at the 85% efficiency WP + $E_T^{miss} < 50$ GeV.

$e\mu$ region:

- Control-region (without m_{ll} criteria) selection with the b -jet veto replaced by a requirement of at least two b -tagged jet at the 85% efficiency WP

The selection cuts of the VRs are further summarized in table 6.1.

Chapter 7

Results

7.1 Measured scale factors

The measured corrections with their relative uncertainties to be applied corresponding to each p_T and $|\eta|$ bin is shown in fig. 7.1. Note that the high p_T region exhibits a clear dependence on pseudorapidity. At small values of $|\eta|$, the scale factors are generally below unity, with values around $0.75 - 0.95$. This indicates that the simulation tends to overestimate the charge-flip contribution in the central detector region. As $|\eta|$ increases, the scale factors gradually rise and become larger than unity, reaching values of approximately $1.2 - 1.7$ in the forward region. This trend suggests that the MC increasingly underestimates the charge-flip probability at higher pseudorapidities, requiring larger corrections to match the observed data.

Overall, the magnitude of the correction grows with $|\eta|$ in the high p_T region, which is consistent with the expected increase in the charge misidentification in the forward region. At low p_T and high $|\eta|$, the uncertainties increase due to the lower event statistics.

The measured corrections are subsequently applied to charge-flip MC. Each charge-flip truth electron found in each MC event is assigned to the measured scale factor in the corresponding $(p_T, |\eta|)$ bin, and the total charge-flip MC event weight is then multiplied that scale factor. Events which are assigned scale factors greater than unity receive an increased weight, while events weight in bins with scale factors below unity are correspondingly reduced. This procedure corrects the simulated charge-flip MC yields in any region.

Scale factors with suppression of photon conversions

As seen in figure 6.8, there is a non-negligible contribution from electrons from photon conversion in every slice of the control region. Reconstruction level variables called AmbiguityType and DFCAA, can be used to suppress this contribution by requiring AmbiguityType = 0 and DFCAA ≤ 0 to every electron. The distributions for Ambiguity type and DFCAA are shown in fig 7.3.

As seen in figs. 7.3(b) and 7.3(d), while these selections reduce the contribution from conversion electrons, they also concurrently suppress the charge-flip events, thereby reducing the available statistics useful for the scale factor measurement.

The resulting scale factor map after applying the conversion veto criteria is illustrated in fig. 7.2. The modified values of corrections in each $(p_T, |\eta|)$ bin are broadly consistent with those without the conversion veto. However, the reduced event yields lead to substantially larger uncertainties. Therefore, the scale factor map derived without the conversion veto criteria is used in the analysis.

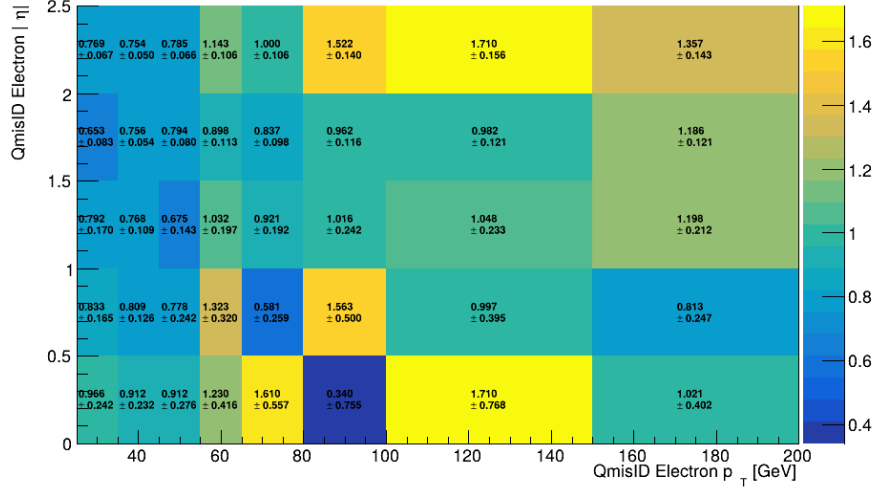


Figure 7.1: Charge-flip scale factor map as a function of the probe-electron transverse momentum (p_T) and pseudorapidity ($|\eta|$). The values shown in each bin correspond to the corrections, with the quoted uncertainties representing the statistical uncertainty of the measurement.

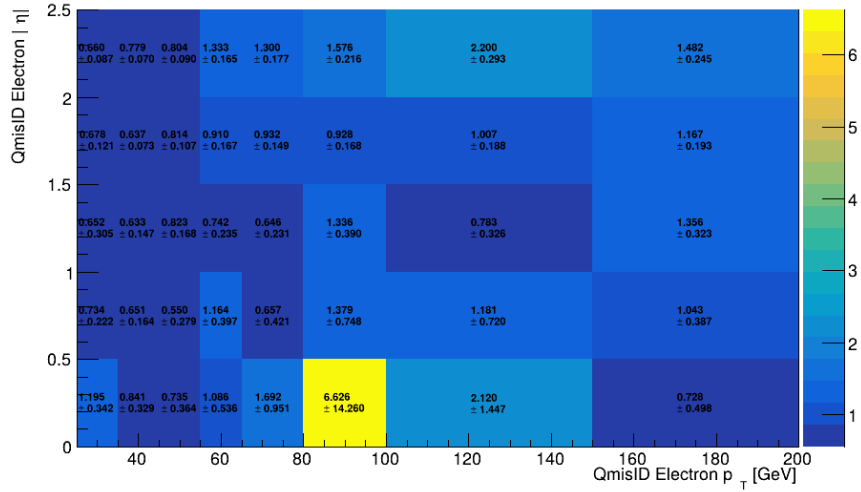
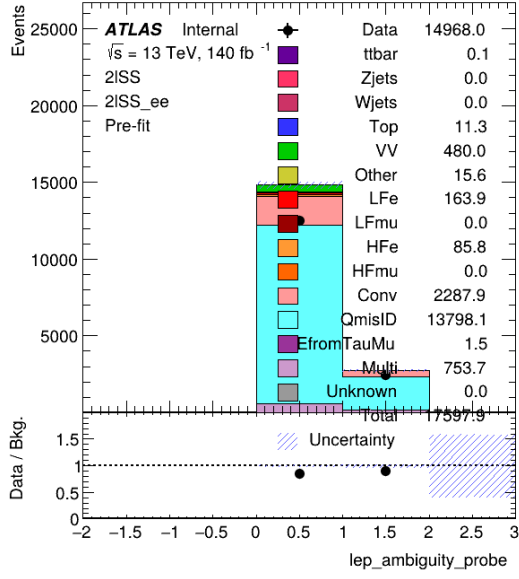
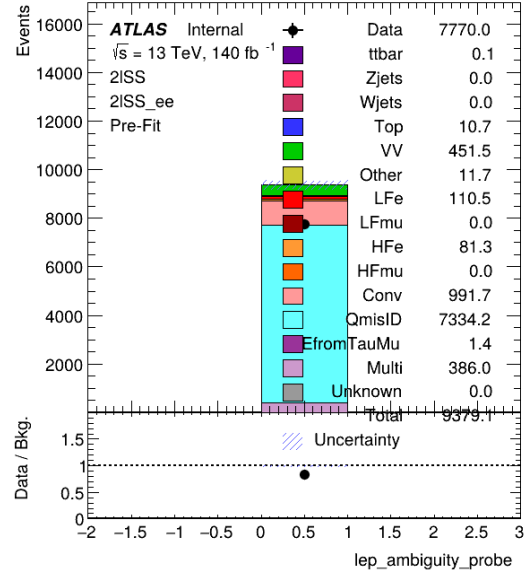


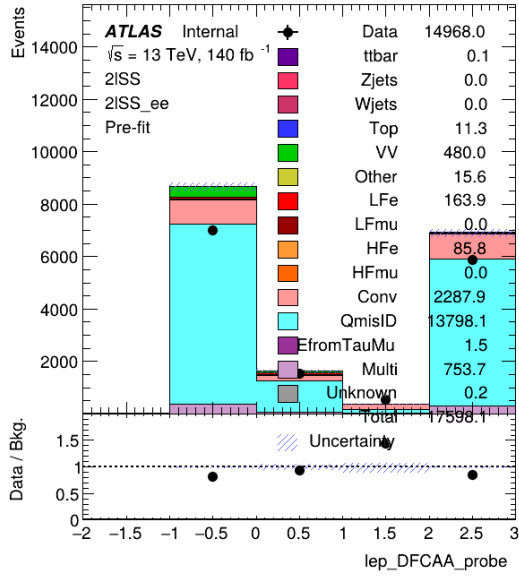
Figure 7.2: Charge-flip scale-factor map as a function of the probe-electron transverse momentum (p_T) and pseudorapidity ($|\eta|$) after applying the conversion-veto criteria. The values shown in each bin correspond to the corrections, with the quoted uncertainties representing the statistical uncertainty of the measurement.



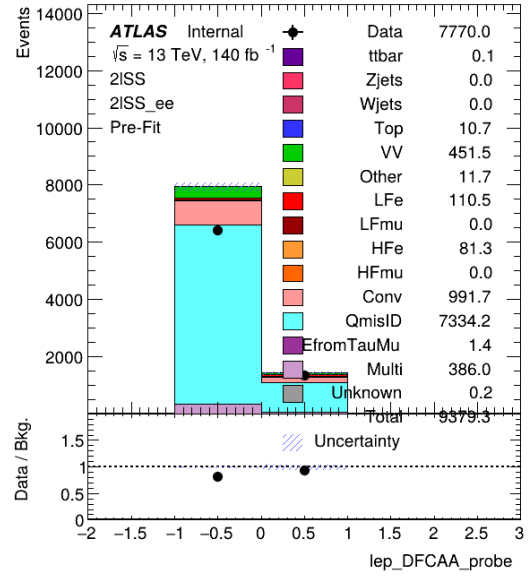
(a)



(b)

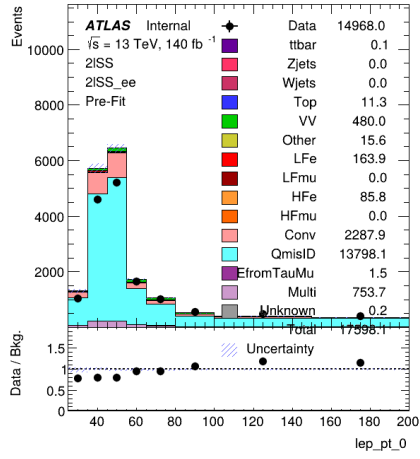


(c)

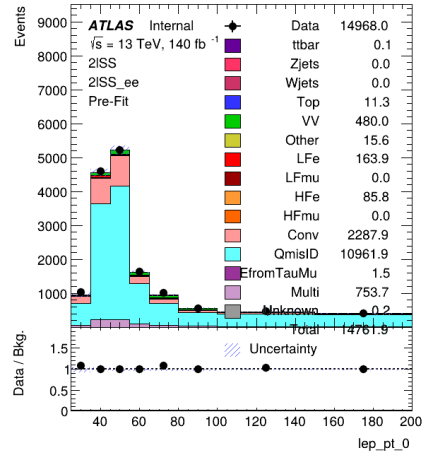


(d)

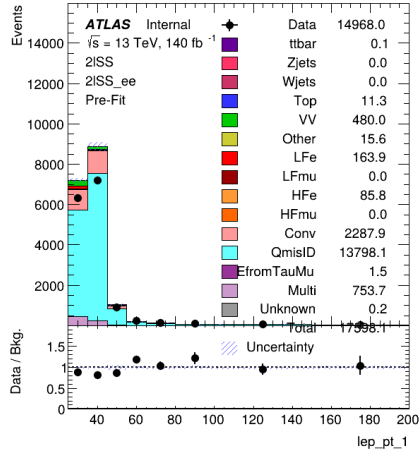
Figure 7.3: Distributions of the photon conversion-veto variables AmbiguityType (a),(b) and DFCAA (c),(d) for probe electrons in the charge-flip control region. Panels (a) and (c) show the distributions before the conversion-veto selection, while panels (b) and (d) show the distributions after the veto criteria.



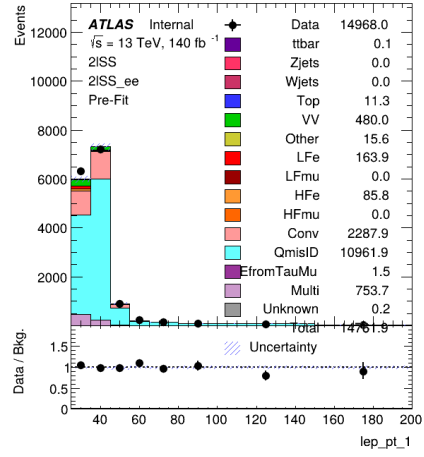
(a)



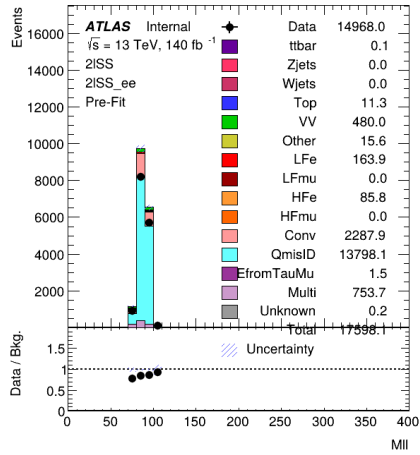
(b)



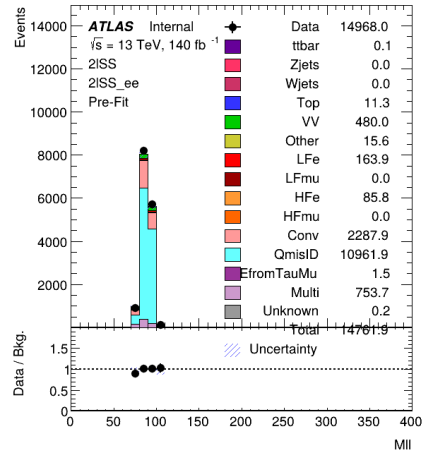
(c)



(d)

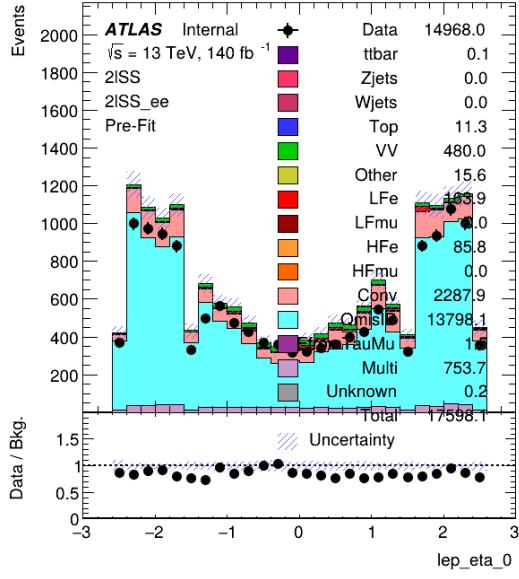


(e)

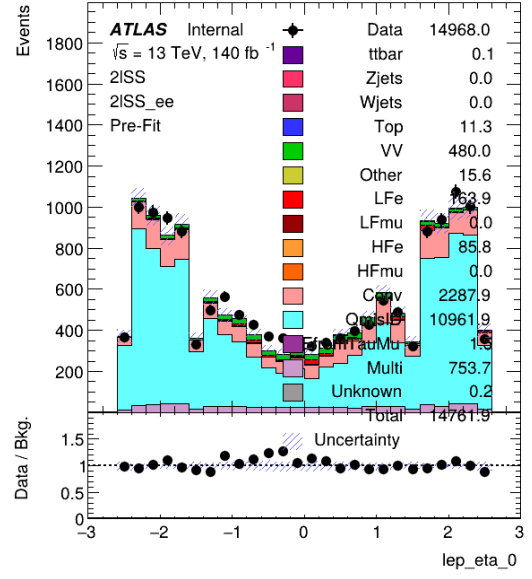


(f)

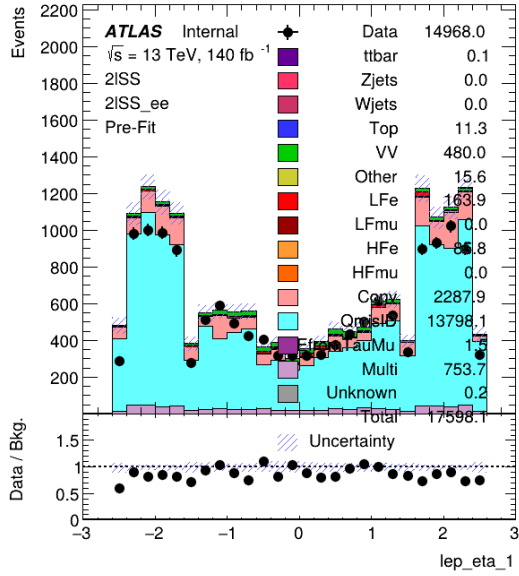
Figure 7.4: Distributions of the leading-electron transverse momentum $p_T^{(0)}$ (a),(b), and subleading-electron transverse momentum $p_T^{(1)}$ (c),(d), and di-electron invariant mass $m_{\ell\ell}$ (e),(f) in the charge-flip control region. The left panels show the distributions before applying the charge-flip corrections, while the right panels show the corresponding distributions after the corrections are applied.



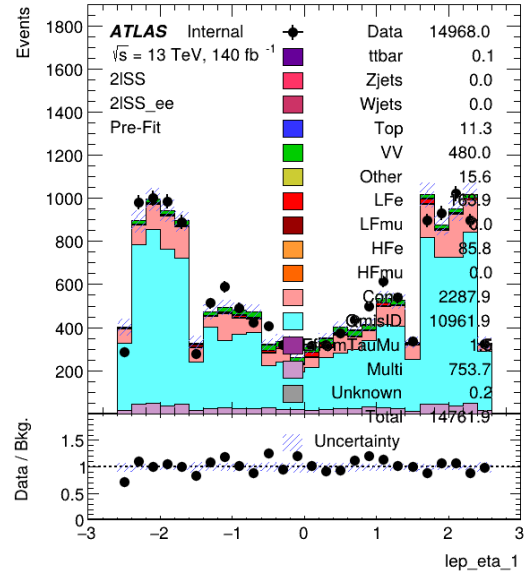
(a)



(b)



(c)



(d)

Figure 7.5: Distributions of the leading-electron pseudorapidity $\eta^{(0)}$ (a),(b), and subleading-electron pseudorapidity $\eta^{(1)}$ (c),(d) in the control region. Distributions before (left) and after (right) the corrections are compared.

7.2 Effect of scale factors in the control region

The impact of the corrections are shown in figs. 7.4, 7.5. It is noticed after applying the corrections, a better agreement between data and the MC is observed in the leading and subleading electron p_T and η distributions, as well as in the di-electron invariant mass distribution (figs. 7.4(b), 7.4(d) and 7.4(f)). This demonstrates that the corrections effectively reduce the mis-modeling of the charge-flip process in MC simulations.

7.3 Effect of scale factors in the validation regions

ee channel

(a) The effect of charge-flip corrections in the first validation region are shown in figs. 7.6, 7.7. The validation region shows generally a good agreement between data and the MC background prediction both before and after the applying the charge-flip corrections. The corrections lead to a modest improvement in the modeling of the leading and subleading electron p_T , pseudorapidity (η), and dielectron invariant mass distributions (m_{ll}). A slight improvement is observed around the Z -boson mass peak and in the forward η region, where charge-flip effects are expected to be more significant. This demonstrates that the correction factors derived in the control region still apply to this VR.

(b) This VR is required to have at least 2 b-jets at 85% WP. The requirement of at least two b-tagged jets significantly alters the composition of the selected events compared to the previous validation region. As expected, the $t\bar{t}$ contribution becomes dominant, while the relative fraction of charge-flip events is substantially reduced ($\sim 14\%$) as seen in figs. 7.8, 7.9. Consequently, the impact of charge-flip corrections to this region is smaller than in a region enriched in Z +jets events. This behavior is visible in the leading and subleading electron p_T , η and invariant mass (m_{ll}) distributions, where the MC is already in reasonable agreement with the data before applying the corrections. After the correction, only small changes are observed in the charge-flip component and in the overall background prediction, resulting in a correspondingly modest impact on the data/MC agreement.

$e\mu$ channel

This VR differs significantly from the control region as well as the other validation region as it further suppresses Z +jets charge-flip events. Consequently, the region is dominated by the $t\bar{t}$ production, while the contribution from charge-flip events remains small since charge mis-identification affects only the electron and not the muon. Only a limited impact from the charge-flip corrections is seen in the figs. 7.10, 7.11. Similar to the ee channel with at least 2 b -jets, the improvement before and after corrections remains small, although still acceptable.

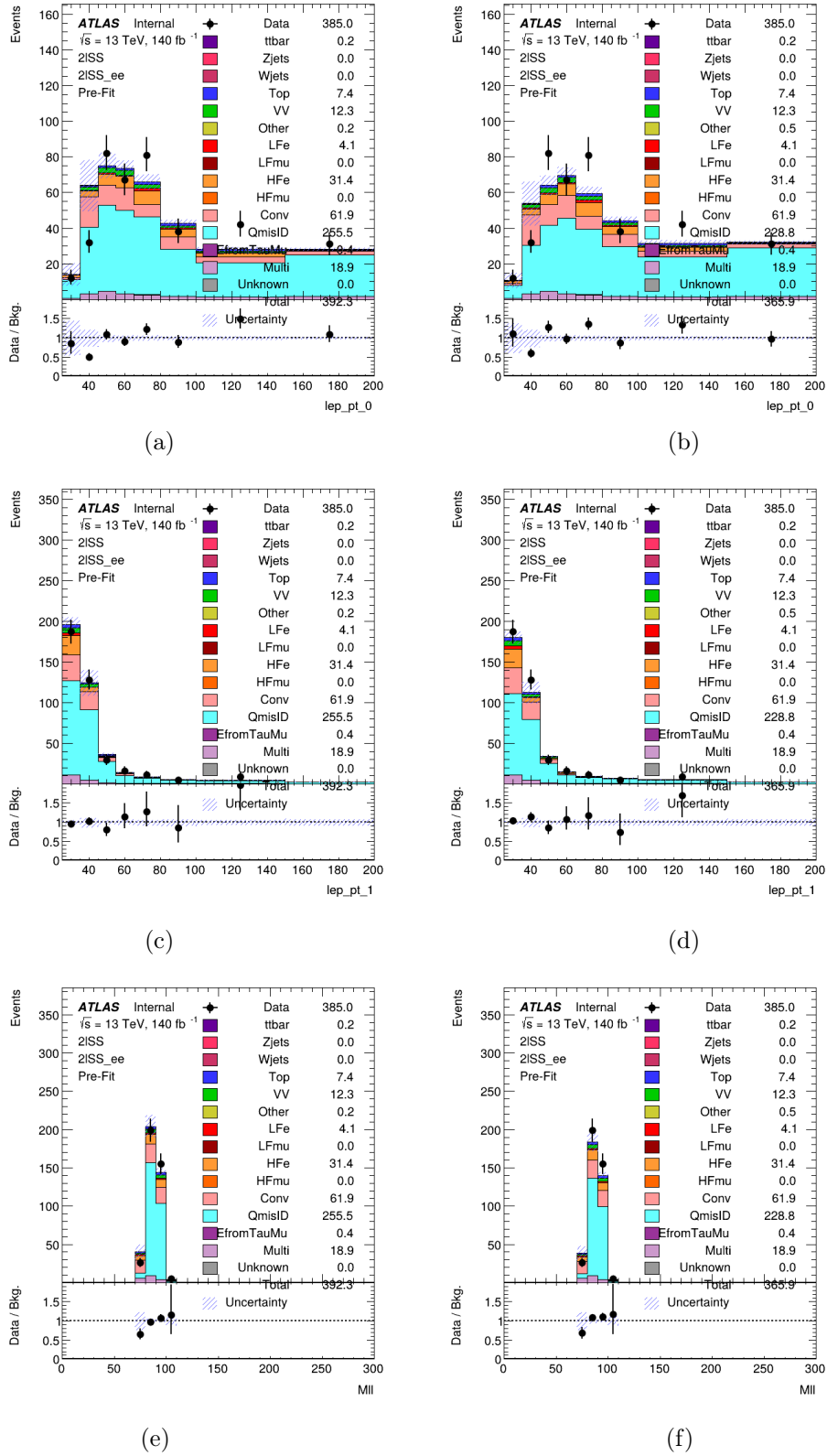
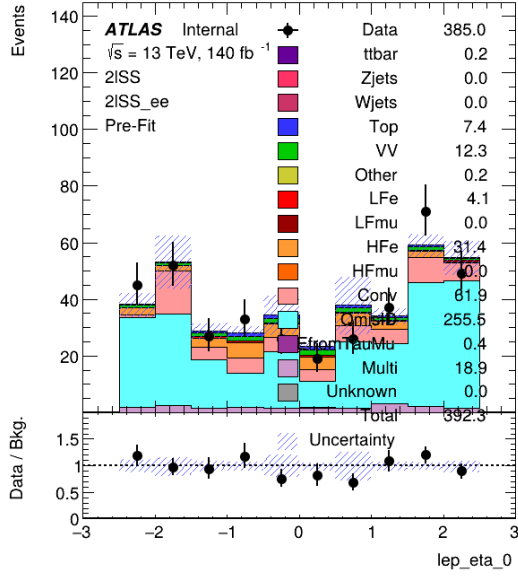
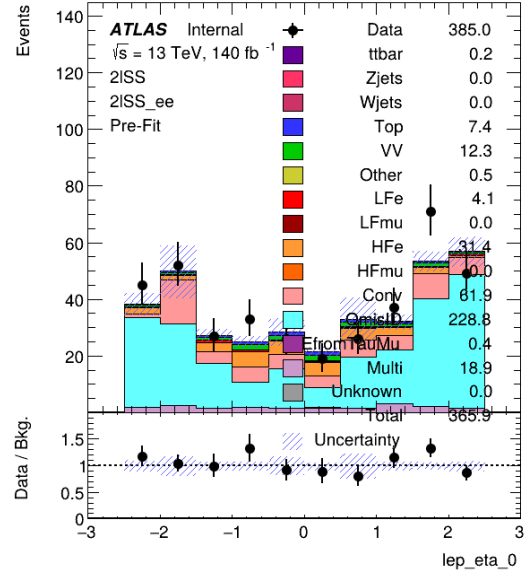


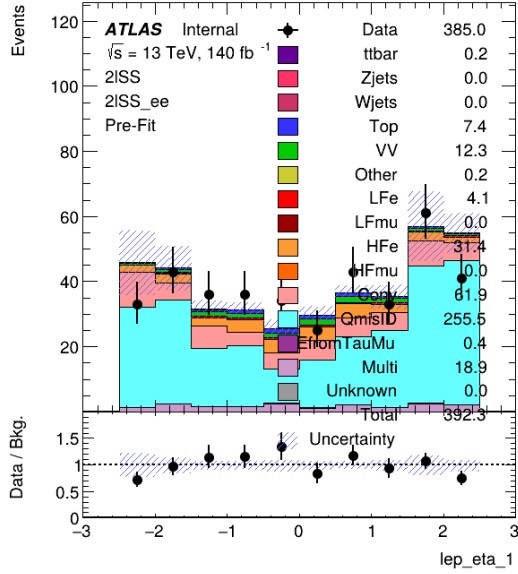
Figure 7.6: Leading-electron transverse momentum ($p_T^{(0)}$) (a),(b) and subleading-electron transverse momentum ($p_T^{(1)}$) (c),(d), and di-electron invariant mass $m_{\ell\ell}$ (e),(f) distributions in the exactly 1 b -jet electron-electron validation region. The left panels show the distributions before applying the charge-flip corrections, while the right panels show the corresponding distributions after the corrections are applied.



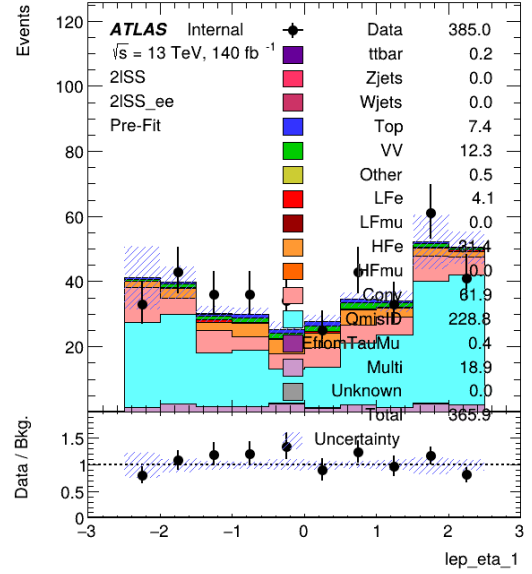
(a)



(b)



(c)



(d)

Figure 7.7: Leading-electron pseudorapidity ($\eta^{(0)}$) (a),(b) and subleading-electron pseudorapidity ($\eta^{(1)}$) (c),(d) with exactly 1 b -jet in the electron-electron validation region. The left panels show the distributions before applying the charge-flip corrections, while the right panels show the corresponding distributions after the corrections are applied.

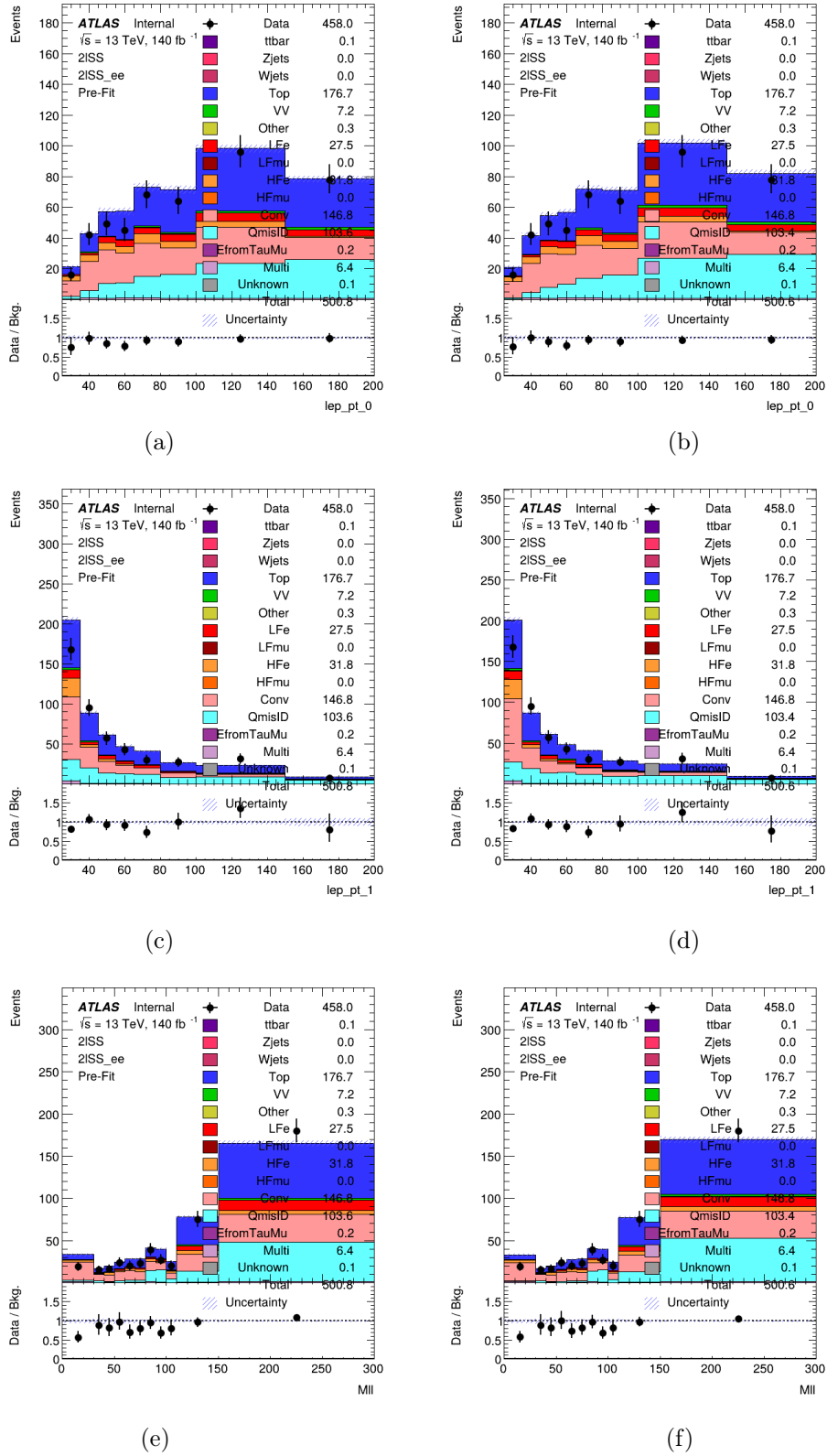
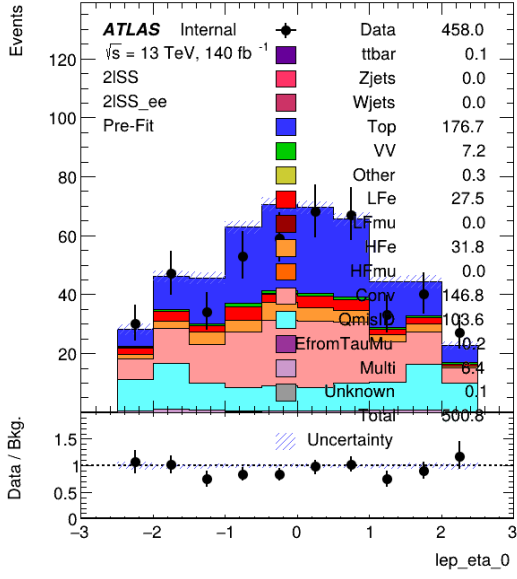
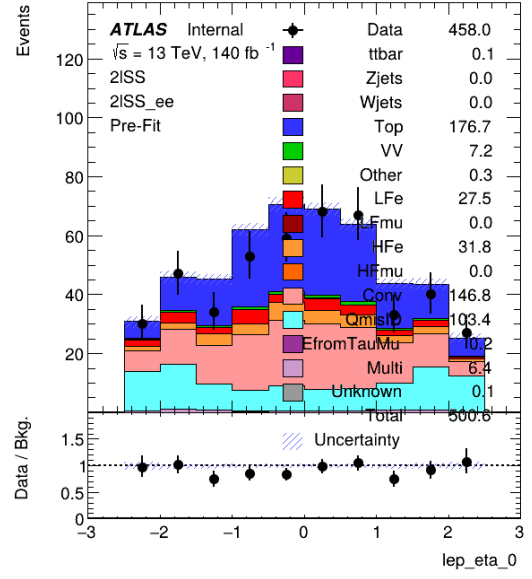


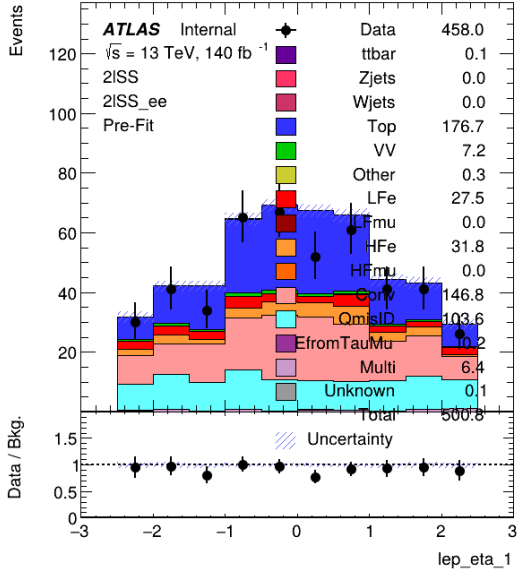
Figure 7.8: Leading-electron transverse momentum ($p_T^{(0)}$) (a),(b) and subleading-electron transverse momentum ($p_T^{(1)}$) (c),(d), and di-electron invariant mass $m_{\ell\ell}$ (e),(f) distributions with at least 2 b -jets in the electron-electron validation region. The left panels show the distributions before applying the charge-flip corrections, while the right panels show the corresponding distributions after the corrections are applied.



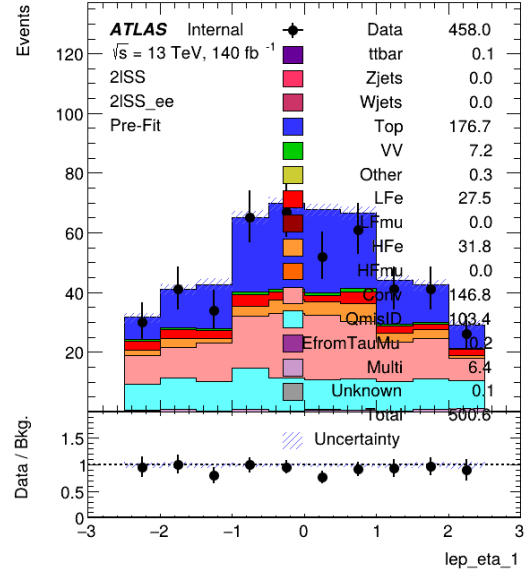
(a)



(b)

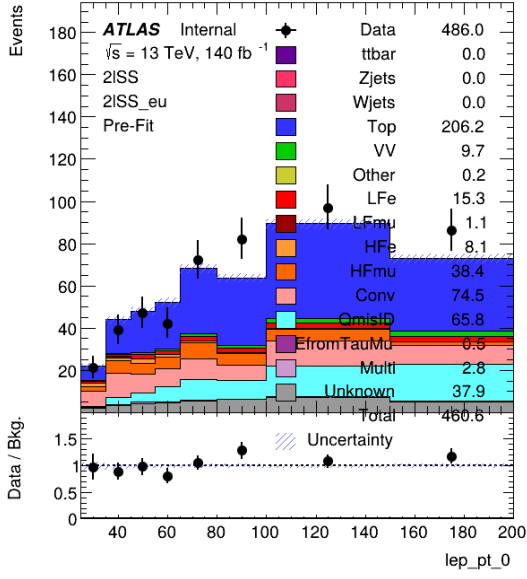


(c)

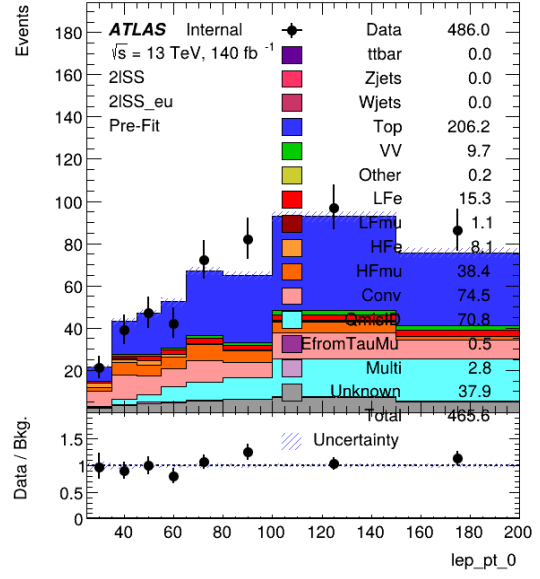


(d)

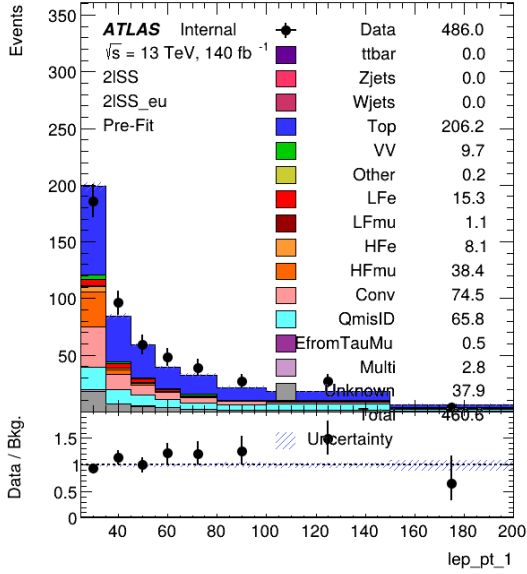
Figure 7.9: Leading-electron pseudorapidity ($\eta^{(0)}$) (a),(b) and subleading-electron pseudorapidity ($\eta^{(1)}$) (c),(d), with at least 2 b -jets in the electron-electron validation region. The left panels show the distributions before applying the charge-flip corrections, while the right panels show the corresponding distributions after the corrections are applied.



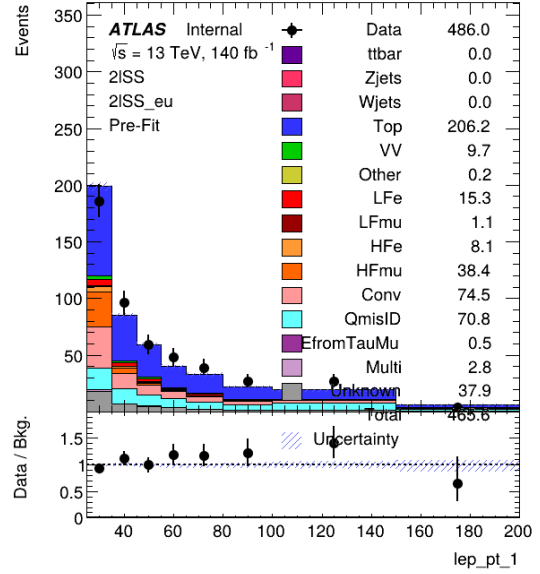
(a)



(b)

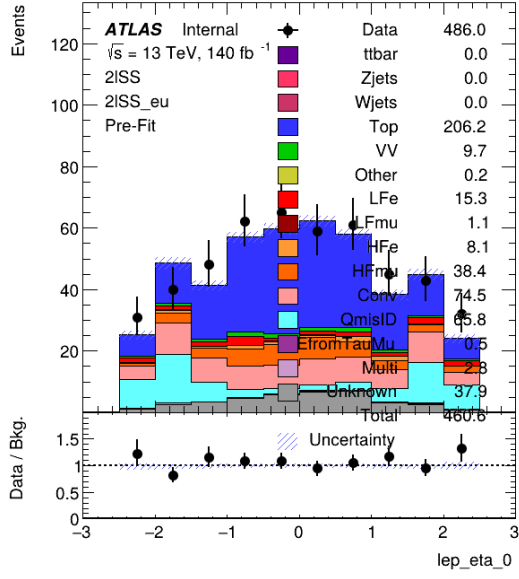


(c)

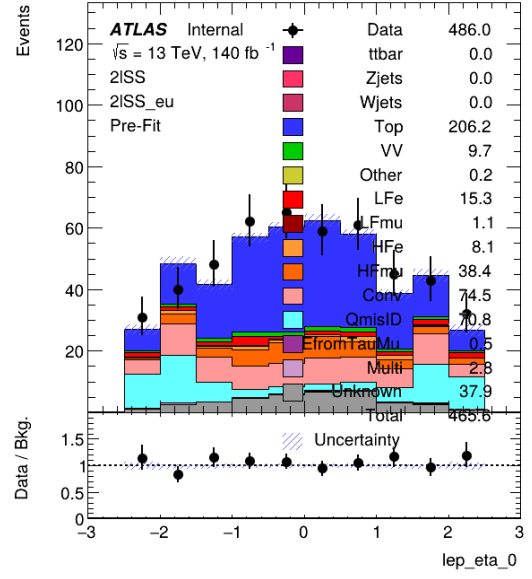


(d)

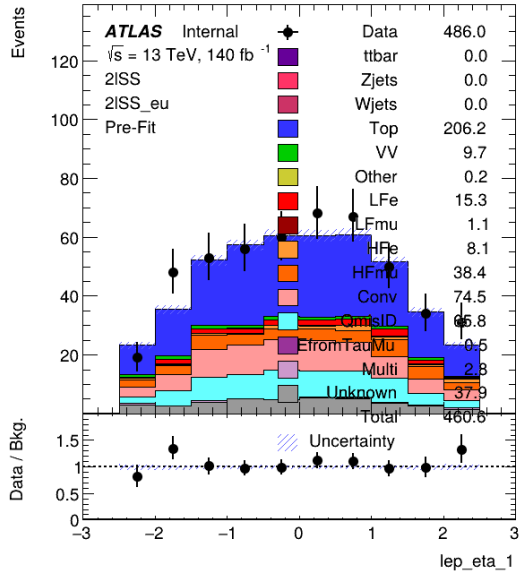
Figure 7.10: Leading-electron transverse momentum ($p_T^{(0)}$) (a),(b) and subleading-muon transverse momentum ($p_T^{(1)}$) (c),(d), and di-electron invariant mass $m_{\ell\ell}$ (e),(f) distributions with at least 2 b -jets in the electron-muon validation region. The left panels show the distributions before applying the charge-flip corrections, while the right panels show the corresponding distributions after the corrections are applied.



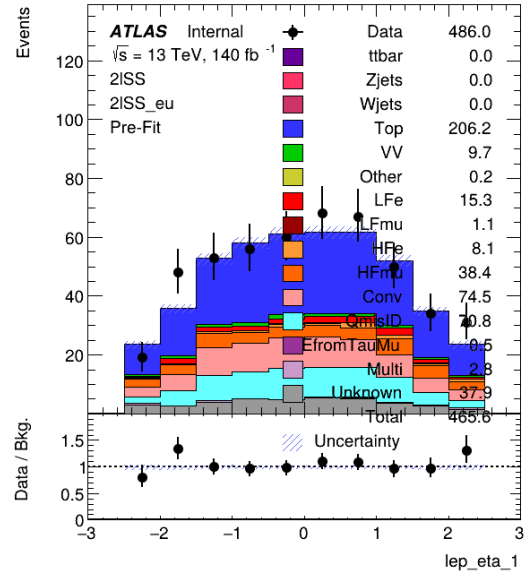
(a)



(b)



(c)



(d)

Figure 7.11: Leading-electron pseudorapidity ($\eta^{(0)}$) (a),(b) and subleading-muon pseudorapidity ($\eta^{(1)}$) (c),(d), with at least 2 b -jets in the electron-muon validation region. The left panels show the distributions before applying the charge-flip corrections, while the right panels show the corresponding distributions after the corrections are applied.

Chapter 8

Summary and outlook

This thesis presents a study of the charge-flip background in the 2-lepton with same charge (2lSS) electron channel using the full Run-2 pp collision data collected at center of mass energy of $\sqrt{s} = 13$ TeV and luminosity $L = 140 \text{ fb}^{-1}$ by the ATLAS detector.

The 2lSS channel is very sensitive charge-flip events, therefore an accurate modeling of this background is crucial for precise SM measurements and searches beyond the SM in this region.

We employ a semi data-driven tag-and-probe method to derive scale factors to correct the charge-flip MC prediction. A charge-flip control region was defined around the Z mass peak, where the dominant contribution comes from Z +jets events.

Scale factors were measured from data as a function of p_T and $|\eta|$ of the probe electron. The derived corrections showed a clear dependence on the charge-flip electron kinematic. Larger corrections were required in high pseudorapidity ($|\eta|$) region. After applying the corrections, an improved data/MC agreement was observed in the charge-flip control region. The most significant improvements were seen in the leading and sub-leading electrons p_T and di-electron invariant mass distributions.

The robustness of the procedure was further evaluated using dedicated validation regions (VR) involving b -tagged jets and electron-muon final states. The di-electron + b -jets VR reveal a small improvement after applying the corrections. The charge-flip purity in the $e\mu$ VR is significantly reduced in both selection criteria. As a result, the data/MC agreement in these region does not change dramatically after the corrections are applied, although it remains consistent.

The charge-flip corrections derived in this work provide an important improvement in the modeling of the mis-reconstruction of electron charge in 2 lepton same sign events. The studies in the validation region also demonstrate that the corrections can be successfully applied in regions with different event topologies and compositions. This indicates that the method used in this thesis can generally be applied in any analysis that features the mis-reconstruction of the electron charge.

The results of this thesis are relevant for future analyses in same sign electron and multilepton final states where charge-flip events can constitute an important reducible background. It is directly applicable to an ongoing multilepton based analysis using the Anomaly Detection (AD) technique [18]. An analogous procedure will be used to measure corrections for the ATLAS run 3 dataset, as well.

In conclusion, this work successfully demonstrates how dedicated measurements of the charge-flip MC corrections can improve the prediction of this important background.

Bibliography

- [1] ATLAS Collaboration. Electron efficiency measurements with the ATLAS detector using 2012 LHC proton–proton collision data. *Eur. Phys. J. C*, 77(3):195, 2017.
- [2] ATLAS Collaboration. The atlas experiment. *Scholarpedia*, 9(10):32147, 2014.
- [3] ATLAS Collaboration. The atlas experiment at the cern large hadron collider. *Journal of Instrumentation*, 3(08):S08003, aug 2008.
- [4] ATLAS Collaboration. Muon reconstruction performance of the ATLAS detector in proton–proton collision data at $\sqrt{s}=13$ TeV. *Eur. Phys. J. C*, 76(5):292, 2016.
- [5] ATLAS Collaboration. Electron reconstruction and identification in the ATLAS experiment using the 2015 and 2016 LHC proton-proton collision data at $\sqrt{s}=13$ TeV. *Eur. Phys. J. C*, 79(8):639, 2019.
- [6] ATLAS Collaboration. The new atlas track reconstruction (newt). *Journal of Physics: Conference Series*, 119(3):032014, jul 2008.
- [7] R Frühwirth. A gaussian-mixture approximation of the bethe–heitleer model of electron energy loss by bremsstrahlung. *Computer Physics Communications*, 154(2):131–142, 2003.
- [8] ATLAS Collaboration. Search for Beyond the Standard Model physics with anomaly detection in multilepton final states in pp collisions at $\sqrt{s}=13$ TeV with the ATLAS detector. *Eur. Phys. J. C*, 86(3):247, 2026.
- [9] Andreas Hoecker, Peter Speckmayer, Joerg Stelzer, Jan Therhaag, Eckhard von Toerne, Helge Voss, M Backes, T Carli, O Cohen, A Christov, et al. Tmva-toolkit for multivariate data analysis. *arXiv preprint physics/0703039*, 2007.
- [10] ATLAS Collaboration. Muon reconstruction and identification efficiency in ATLAS using the full Run 2 pp collision data set at $\sqrt{s}=13$ TeV. *Eur. Phys. J. C*, 81(7):578, 2021.
- [11] ATLAS Collaboration. Configuration and performance of the ATLAS b-jet triggers in Run 2. *Eur. Phys. J. C*, 81(12):1087, 2021.
- [12] ATLAS Collaboration. Performance of missing transverse momentum reconstruction with the ATLAS detector using proton-proton collisions at $\sqrt{s}=13$ TeV. *Eur. Phys. J. C*, 78(11):903, 2018.
- [13] Torbjorn Sjostrand, Stephen Mrenna, and Peter Z. Skands. A Brief Introduction to PYTHIA 8.1. *Comput. Phys. Commun.*, 178:852–867, 2008.
- [14] ATLAS Collaboration. The ATLAS Simulation Infrastructure. *Eur. Phys. J. C*, 70:823–874, 2010.
- [15] Geant4—a simulation toolkit. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 506(3):250–303, 2003.
- [16] W. Lukas. Fast Simulation for ATLAS: Atfast-II and ISF. *J. Phys. Conf. Ser.*, 396:022031, 2012.
- [17] G. Cowan. *Statistical data analysis*. 1998.

- [18] ATLAS Collaboration. Search for Beyond the Standard Model physics with anomaly detection in multilepton final states in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector. *Eur. Phys. J. C*, 86(3):247, 2026.